Sampling Design for Spatial Cluster Randomized Trials Under Heterogeneous Incidence

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Introduction

Cluster randomized trials (CRTs) randomize intact groups, such as clinics, schools, or geographic areas, rather than individuals, and are the standard design when an intervention is delivered at the group level or when individual randomization would create unacceptable contamination between arms [1, 2]. A central assumption underlying standard CRT analysis is the no-interference component of the stable unit treatment value assumption (SUTVA): a cluster's outcome must not depend on which other clusters were assigned to treatment [3]. This assumption, borrowed from individually randomized trial theory, can fail once clusters are spatially arranged, and so can the weaker assumption of partial interference, under which outcomes may depend on assignments within prespecified groups but not across them [4]. Tobler's first law of geography, "everything is related to everything else, but near things are more related than distant things" [5], implies that an intervention delivered in one location can plausibly influence outcomes in geographically adjacent locations through shared populations, service networks, information diffusion, or environmental pathways. Documented examples of this kind of spatial spillover span public health and development economics: in a trial of insecticide-treated bed nets in northern Ghana, child mortality was lower in control villages close to treated villages than in

control villages farther away [6], and a school-based deworming trial in Kenya found health and attendance benefits among untreated children near treated schools [7]. When control clusters benefit from nearby treatment in this way, an unadjusted comparison of treated and control clusters may understate the direct effect, and how much contamination occurs can depend on the spatial arrangement of treated and control clusters, not only on the number of clusters allocated to each arm.

The design-stage gap in the spatial interference literature

The methodological literature addressing spatial interference in CRTs has concentrated on analysis-stage correction (Chapter 1). Spatial autocorrelation statistics such as Moran's I [8] detect residual spatial dependence once outcomes have been observed, and spatial regression models, such as the spatial lag model [9] estimated by maximum likelihood [10] and implemented in software [11], allow an analyst to separate a direct treatment effect from a spillover effect after the fact. In their systematic review of spatial analysis methods for cluster randomized trials, Jarvis and colleagues placed trials that used spatial separation or buffers outside the primary review of statistical methods [12]. The literature therefore offers much more guidance on analyzing spatial effects than on how the choice of treatment assignment itself, made before any data are collected, might reduce the problem.

Design-stage tools do exist, but they were largely developed for a different purpose. Restricted and covariate-constrained randomization address the baseline imbalance that a limited number of clusters can produce [13, 14]; Crisp and colleagues noted that their extension, which selects clusters from a larger eligible set, also increases spatial dispersion among the selected clusters [15]. Their dispersion criterion concerns which clusters are selected rather than how they are assigned to arms, and balancing a covariate across arms does not by itself determine whether treated and control clusters are spatially interspersed in a way that induces contamination. Work that addresses the problem studied here more directly includes randomized saturation designs, which

assign different fractions of units treated to different groups so that direct and spillover effects can be jointly identified [16]; the selection of whole clusters for inclusion or exclusion in place of a "fried-egg" design that omits each cluster's outer area from analysis [17]; and recent theoretical designs under spatial, network and cross-cluster interference and spatio-temporal heterogeneity [18, 19, 20, 21]. This chapter complements this work with a simulation comparison of concrete assignment strategies that a trial team could implement, under heterogeneous baseline incidence and a range of spatial dependence and spillover conditions.

Extending a small-grid proof of concept

Chapter 2 examined block-stratified allocation as a practical way to obtain reasonably good estimation while limiting poor-allocation risk in its studied settings; this chapter evaluates both criteria across a broader set of designs under heterogeneous incidence. We retain its simple random assignment benchmark and strict alternating allocation, called Checkerboard here, while adding incidence balancing, targeting, saturation and isolation strategies. The grid comparison increases the number of clusters from 8–12 to 100, considers queen as well as rook neighbors, and includes heterogeneous baseline incidence in the outcome model. These changes extend the design question rather than provide a direct numerical replication of Chapter 2. In Chapter 2, β denotes the intervention effect; here that effect is τ, while β denotes the coefficient of the incidence covariate. Our observed-incidence application subsequently uses the North Carolina (NC) community college geography and a continuous education outcome, without assuming that historical sudden unexpected death (SUD) incidence predicts baseline education performance.

Aims

This chapter has three aims: compare assignment designs by mean squared error (MSE), bias and confidence interval coverage; assess how conclusions depend on incidence heterogeneity, spatial dependence, spillover regime and treatment-effect magnitude; and assess both average

accuracy and allocation-specific downside for planning a SUD-prevention education pilot on an observed North Carolina geography. The detailed grid results retain a six-design presentation, with the two related variants documented in the appendix. A main-text benchmark includes all eight strategies and simple random sampling (SRS). The completed application evaluates all nine strategies separately for each observed planning year. The application simulates outcomes and assignments; it does not report outcomes from an implemented intervention.

Methods

Spatial structure and outcome model

We simulated a 10×10 regular grid of 100 clusters under two definitions of adjacency. Under queen contiguity, two clusters are neighbors if they share an edge or a corner; under rook contiguity, they must share an edge. Queen contiguity is the primary analysis in this chapter. Rook contiguity is a sensitivity analysis and the case that continues Chapter 2, whose grids used rook neighbors. For each definition, the binary adjacency matrix was row-standardized to form the spatial weights matrix W, so that WZ gives, for each cluster, the proportion of its neighbors that are treated, and WY gives the average outcome among its neighbors. Row-standardization expresses each cluster's spillover exposure as an average over its own neighborhood rather than a raw sum, which would otherwise depend on how many neighbors the cluster has. Within a scenario, the same neighbor type defines W in the outcome model, in the Isolation Buffer design, and in the fitted model.

Cluster-level outcomes Y were generated from a spatial lag model with a spillover term,

$$Y = (I - \rho W)^{-1} \left[\tau Z + S(Z) + \beta X + \epsilon\right], \qquad \epsilon \sim N(0, \sigma^2 I), \tag{1}$$

where $Z \in \{0,1\}^{100}$ is the treatment assignment produced by the design under evaluation, τ

is the direct treatment effect of interest, $S(Z)$ is the spillover term defined in Equation 2, ρ is the spatial autoregressive coefficient governing dependence in the outcome beyond that induced by treatment and spillover, X is the baseline incidence covariate with coefficient $\beta = 1$, and $\sigma = 1$. The spatial multiplier $(I - \rho W)^{-1}$ propagates each cluster's shocks through the network of neighbors. We tested $\rho \in \{0, 0.01, 0.20, 0.50\}$. When $\rho = 0$ the multiplier is the identity and the model is a linear regression with independent errors, $Y = \tau Z + S(Z) + \beta X + \epsilon$, in which spatial structure enters only through the spillover term; larger values of ρ induce progressively stronger spatial clustering in the outcome.

We evaluated two spillover regimes,

$$S(Z) = \gamma\, WZ \quad \text{(both arms)}, \qquad S(Z) = \gamma\, WZ \circ (1 - Z) \quad \text{(control only)}, \tag{2}$$

where $\circ$ denotes the elementwise product. Under the control-only regime, only untreated clusters respond to the treatment status of their neighbors, which represents classical contamination: an untreated cluster's outcome rises with the share of its neighbors that are treated. Under the both-arms regime, every cluster responds to its neighbors' treatment regardless of its own assignment, so neighboring treatment adds to a treated cluster's own effect. We tested four spillover magnitudes, $\gamma \in \{0.5, 0.6, 0.7, 0.8\}$, and five magnitudes of the direct effect, $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$, with $\tau = 1$ as the primary scenario throughout.

Heterogeneous baseline incidence

Chapter 2 assumed a spatially uniform baseline outcome risk. Cluster-level health outcomes typically vary across space, and the designs studied here differ in whether and how they use that variation. We generated the baseline incidence covariate X, with values in $(0, 1)$, under three mechanisms. Under the independent (iid) mechanism, each X_i is drawn from a Uniform$(0, 1)$ distribution, with no spatial structure. Under the spatially autocorrelated mechanism, $X =$

$\Phi\{(I - \rho_X W_Q)^{-1} u\}$, where $u \sim N(0, I)$, Φ is the standard normal distribution function applied elementwise, W_Q is the queen weights matrix, and $\rho_X \in \{0.20, 0.50\}$. Under the Poisson mechanism, a spatially autocorrelated log-rate surface, $\log \lambda = \log(35/100{,}000) + (I - \rho_X W_Q)^{-1} u$ with $\rho_X \in \{0.20, 0.50\}$, generates cluster death counts $C_i \sim \text{Poisson}(\lambda_i P)$, with an equal population of $P = 100{,}000$ in every cluster, so that a cluster at the base rate expects 35 deaths. The observed rates C_i/P are then rank-normalized to $X_i = (r_i - 0.5)/N$, where r_i is the rank of cluster i's rate among the $N = 100$ clusters and tied rates share their average rank. Ties remain at this population size: the Poisson surfaces used here have between 60 and 76 distinct values among the 100 clusters. The base rate of 35 per 100,000 was chosen to be representative of published SUD incidence: it lies between the SUD incidence rates that Mirzaei and colleagues estimated for US men (49.3 per 100,000) and women (21.7 per 100,000) aged 20–64, close to their average [22]. It is not the observed rate of the county data described in the Application section, which come from a different source, age range, period, and case definition. The incidence filters always use the queen weights W_Q, whichever neighbor type the scenario uses, so the incidence surfaces are identical under rook and queen contiguity. The iid mechanism and the two strengths of each of the other two mechanisms give five incidence configurations.

For each configuration we generated $K = 10$ incidence surfaces, $X_1, \ldots, X_{10}$, each from its own seed. The same ten surfaces are used by every scenario in that configuration, across neighbor types, outcome-model parameters, designs, and values of τ; the design draws and the outcome noise are shared more selectively, as described under Simulation design, resampling, and metrics. Each design is applied to the surface that generates the outcome: assignments for surface k are drawn from X_k, and the outcome and the fitted model for those assignments use X_k as well. This assumes that the incidence observed when a trial is planned is the incidence that drives its outcome (the matched-surface assumption). The target of each scenario is performance averaged over incidence surfaces; for example, $\text{MSE} = E_X E_{Z|X} E_\epsilon[(\hat{\tau} - \tau)^2]$, with X drawn from the configuration's generator, Z from the design given X, and $\epsilon \sim N(0, I)$. Bias, coverage, and

power are defined in the same way.

Treatment assignment designs

We implemented eight treatment assignment designs and present six of them as the primary comparison set in this chapter. The other two, Saturation Quadrants and Balanced Halves, each share the mechanism of a retained design, and in an eight-design comparison at $\tau = 1$ neither could be distinguished from its counterpart under either neighbor type (Appendix A1). Saturation Quadrants assigns the saturations 20%, 40%, 60%, and 80% to the four quadrants in random order, without reference to incidence; its retained counterpart, Incidence-Guided Saturation Quadrants, assigns the higher saturations to the quadrants with higher mean incidence (Table 1). We shorten the second name to Incidence-Guided Saturation in the rest of this chapter, as in the figures. Balanced Halves divides clusters into two halves by incidence rank and treats 25 at random in each; its retained counterpart, Balanced Quartiles, uses four strata. In each pair the excluded design had the marginally lower mean squared error or average rank under at least one neighbor type, so the choice of representative does not rest on performance. The two saturation designs performed similarly, so incidence guidance lets a trial direct its highest treatment saturations to the highest-incidence quadrants with little or no loss of estimation accuracy; the accuracy advantage of both designs over SRS, which arises under control-only spillover (see SRS benchmark and allocation-specific risk), comes from the saturation structure they share. Both variants are included in the full benchmark with SRS and in the application.

Table 1 describes the six retained designs, and Figure 1 shows one assignment from each on a single illustrative incidence surface. Checkerboard is the strict block-stratified allocation of Chapter 2, in which no two rook neighbors share an arm. Chapter 2's Block Stratified Sampling design also admitted relaxed allocations on its 3×3 grid, where the arms cannot be balanced; on its 2×4 and 3×4 grids, as on the 10×10 grid here, the strict allocations are the two checkerboard patterns. We use the two names interchangeably for this allocation and use Checkerboard in the

rest of this chapter. Checkerboard is the only deterministic design: it gives the same assignment in every draw, although each of its fits still receives its own outcome noise. Every other design was drawn independently 25 times for each incidence surface. Designs that rank clusters by incidence break ties at random, afresh for each draw. High Incidence Focus is random only through that tie-breaking; under the iid and spatial mechanisms incidence values are distinct, so it gives the same assignment in all 25 draws on a surface, and its assignment changes only from one surface to the next. Isolation Buffer builds its assignment from the neighbor type of the scenario, and it treats 38 clusters on average under rook contiguity and 21 under queen contiguity. Every other design treats exactly 50.

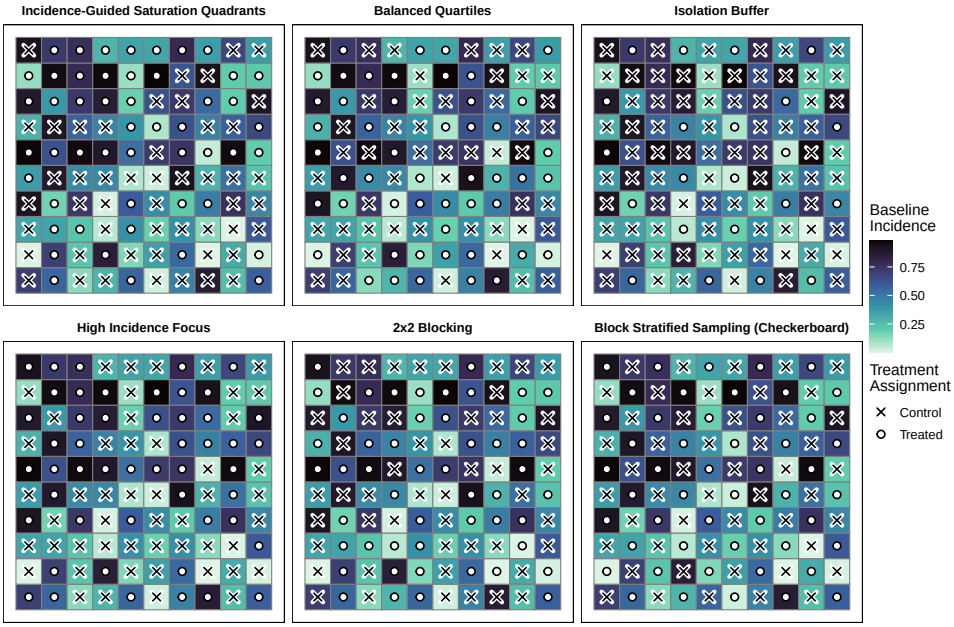


Figure 1: One sample assignment from each of the six designs on the 10×10 grid, with panels in the order of Table 1. All six assignments use a single illustrative incidence surface of 100 independent Uniform$(0, 1)$ values generated with a fixed seed; it is not one of the simulation's incidence surfaces. Tile fill shows baseline incidence (darker is higher), a ring marks a treated cluster, and an X marks a control. Isolation Buffer is shown under queen contiguity. The Checkerboard panel is labeled Block Stratified Sampling (Checkerboard), the Chapter 2 name for this allocation (see Treatment assignment designs).

Estimation

We estimated τ with two maximum likelihood spatial lag models, both fit to every simulated outcome and both including an intercept. The primary, oracle model has the spatial lag ρWY

Table 1: The six treatment assignment designs compared in this chapter, ordered by mean squared error under queen contiguity at $\tau = 1$, pooled over the five incidence configurations (Table 2). A cluster's incidence rank is its rank by the baseline incidence surface X_k, with ties broken at random in every draw. Clusters treated is the number of clusters treated per assignment; for Isolation Buffer, which depends on the neighbor type, it is the mean over its assignments under each neighbor type.

Design	Assignment rule	Clusters treated	Deterministic
Incidence-Guided Saturation	The grid is divided into its four 5×5 coordinate quadrants. Quadrants are ranked by mean incidence and given saturations of 80%, 60%, 40%, and 20% from highest to lowest mean; within each quadrant, that share of its 25 clusters is treated at random	50	No
Balanced Quartiles	Clusters are divided by incidence rank into four strata of 25. Twelve clusters are treated at random in each stratum, and one more in each of two randomly chosen strata	50	No
Isolation Buffer	Clusters are treated one at a time, each chosen at random from those that are neither treated nor adjacent to a treated cluster, until none remain (a greedy random maximal independent set); no two treated clusters are neighbors	38 rook, 21 queen (mean)	No
High Incidence Focus	The 50 clusters with the highest incidence rank are treated	50	No
2×2 Blocking	The grid is divided into 25 non-overlapping 2×2 blocks; two of the four clusters in each block are treated at random	50	No
Checkerboard (strict Block Stratified Sampling allocation in Chapter 2)	Clusters whose grid coordinates (x, y) have an odd sum $x + y$ are treated, so treatment alternates from cluster to cluster; under rook contiguity every neighbor of a treated cluster is a control, and every neighbor of a control is treated	50	Yes

and the covariates Z, Spill $= S(Z)$, and X_k, so it matches the outcome model in Equation 1, including the true regime-specific spillover covariate from Equation 2. The oracle model assumes that the analyst knows the form of the spillover mechanism and has the exact exposure term, so its results are a best case for each design; we return to this assumption as a limitation. The non-oracle model, a sensitivity analysis, has the covariates Z and X_k only and omits the spillover term. Both models are fit to the same outcomes.

We fit both models with our own implementation of the maximum likelihood estimator of the spatial lag model [23], the same estimator that the `lagsarlm` function of the R package

`spatialreg` computes with `method = "eigen"`. It maximizes the concentrated log-likelihood over ρ, computing the log-determinant from the eigenvalues of W [10], estimates the remaining coefficients by least squares given $\hat{\rho}$, and computes standard errors from the analytic information matrix. On 5,120 validation fits it agreed with `lagsarlm` to within 9.7×10^{-8} in $\hat{\tau}$ and to a relative difference of 1.1×10^{-8} in its standard error, at roughly 1% of the computing time; we also refit a 1% sample of the full simulation with `lagsarlm` itself as a check. Confidence intervals are 95% Wald intervals, $\hat{\tau} \pm 1.96\,\widehat{\mathrm{SE}}$, where $\widehat{\mathrm{SE}}$ is the estimated standard error.

Checkerboard illustrates why interspersion alone cannot guarantee precision. Under rook contiguity on the rectangular grid, $WZ = 1 - Z$, so treatment and spillover cannot be separately identified; the fitted coefficient on treatment estimates $\tau - \gamma$. Under queen contiguity, treatment is identified but control-only spillover is weakly separated from treatment. We retain Checkerboard in the main design comparison and flag aliased estimates. Exact matrix identities, boundary information and the rare Isolation Buffer aliases are detailed in Appendix A7. These diagnostics concern the models implemented here.

Simulation design, resampling, and metrics

The full simulation crossed five values of τ, five incidence configurations, two neighbor types, four values of ρ, four values of γ, two spillover regimes, and eight designs, for $5 \times 5 \times 2 \times 4 \times 4 \times 2 \times 8 =$ 12,800 scenarios, of which the six designs reported in this chapter account for 9,600. We call a combination of incidence configuration, neighbor type, ρ, γ, and spillover regime a block; there are $5 \times 2 \times 4 \times 4 \times 2 = 320$ blocks, 160 for each neighbor type. At the primary scenario, $\tau = 1$, the six designs give 1,920 scenarios, and every block contains all six.

Within each scenario, we drew $J = 25$ assignments from each of the $K = 10$ incidence surfaces, for 250 fits per scenario, and gave every fit its own noise vector, so that fit (k, j) has outcome $Y_{kj} = (I - \rho W)^{-1}[\tau Z_{kj} + S(Z_{kj}) + \beta X_k + \epsilon_{kj}]$; Checkerboard's 250 fits share one assignment but have 250 distinct noise vectors. The ten surfaces are independent, so the surface is the primary

sampling unit. Each estimator was fit 3.2 million times. SRS, which assigns exactly 50 of the 100 clusters to treatment by complete randomization, ignoring incidence and geography, was subsequently run with the same matched surfaces, noise keys and 250-fit structure, bringing the complete benchmark to 14,400 scenarios and 3.6 million fits per estimator; the original eight-design results were unchanged.

Every random draw was seeded from a key that names the draw, so the results do not depend on the order in which scenarios are run, and the keys give common random numbers: within a block, all designs share the same noise vectors and incidence surfaces; the five values of τ reuse the same assignments and noise; and the oracle and non-oracle models are fit to the same outcomes. Blocks within an incidence configuration share its ten surfaces.

For each scenario and estimator, we computed bias, the standard deviation (SD) of $\hat{\tau}$, MSE, coverage of the 95% interval, and power, the fraction of fits whose interval lies above zero (a one-sided test of $\tau = 0$ at level 0.025). For both estimators, every one of the 250 fits in every scenario was valid, with a finite estimate and standard error, and the fitting routine raised no warning other than the aliasing flags described above. Monte Carlo standard errors come from the surface-level metrics: with m_k a metric computed on the 25 fits from surface k, $\mathrm{SE}(m) = \mathrm{sd}(m_1, \dots, m_{10})/\sqrt{10}$, and, for metrics pooled over the five incidence configurations, from the 50 independent configuration-by-surface means. Appendix A2 gives the parameter grid, the seed keys, and the exact formulas for the metrics, their standard errors, and paired differences between designs.

We compared designs within each neighbor type with a Friedman rank-sum test on scenario MSE [24], treating the 160 blocks as repeated measures across the six designs. At $\tau = 1$ we followed it with Nemenyi post-hoc comparisons [25, 26] and paired Wilcoxon signed-rank tests [27] with Holm's adjustment for multiple comparisons [28] for all 15 pairs of designs, and we repeated the Friedman test at each other value of τ. Blocks within an incidence configuration

share their surfaces, so these tests treat dependent blocks as independent, which overstates the information about variation from surface to surface. For each pair of designs adjacent in the mean-MSE order, we therefore also ran a paired Wilcoxon signed-rank test over 50 independent units, the 5 configurations times 10 surfaces, with each unit's MSE averaged over that surface's blocks, and a t_{49} interval for the mean difference. Because the values of τ share assignments and noise, Friedman tests at different values of τ are not independent confirmations of one another.

We report results separately for each neighbor type, with queen contiguity primary, and no number is averaged over the two. Unless stated otherwise, every number in the Results is pooled over the five incidence configurations, and every table and figure caption states whether it pools the configurations or shows them separately. We report the ranking within each configuration separately and show the split wherever a conclusion changes.

Results

Design performance under queen contiguity

At the primary scenario, $\tau = 1$, under queen contiguity and pooled over the five incidence configurations, Incidence-Guided Saturation had the lowest mean squared error (MSE = 0.091), followed by Balanced Quartiles (0.131), Isolation Buffer (0.160), High Incidence Focus (0.245), 2×2 Blocking (0.319), and Checkerboard (1.087) (Table 2, Figure 2). Coverage of the 95% intervals was between 0.936 and 0.941 for all six designs (Figure 11 in Appendix A5). Power to reject $\tau = 0$ was 0.930 for Incidence-Guided Saturation, 0.827 for Balanced Quartiles, 0.731 for Isolation Buffer, 0.642 for 2×2 Blocking, 0.610 for High Incidence Focus, and 0.547 for Checkerboard.

Incidence-Guided Saturation also led block by block. Of the 160 queen blocks at $\tau = 1$, Incidence-Guided Saturation had the lowest MSE in 95 (59%), Balanced Quartiles in 58 (36%), and Isolation Buffer in 7 (4%); no other design had the lowest MSE in any block.

Table 2: Performance of the six designs at the primary scenario, $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages the 160 scenarios of one design and neighbor type (5 incidence configurations × 4 values of ρ × 4 values of γ × 2 spillover regimes), each summarizing 250 fits of the oracle model. Rows in both panels are in order of mean squared error (MSE) under queen contiguity. Monte Carlo standard errors of MSE, in parentheses, are computed from the 50 configuration-by-surface means (Methods); those of bias, coverage, and power are all below 0.007. Average rank is the mean of each design's within-block MSE rank among the six designs over the 160 blocks (1 = lowest MSE). Clusters treated is the mean number of clusters treated per assignment. Under rook contiguity, τ is not identified for Checkerboard ($WZ = 1 - Z$); its estimates are kept and flagged, and its coefficient on Z estimates $\tau - \gamma$.

Design	MSE (SE)	Bias	Coverage	Power	Average rank	Clusters treated
Queen contiguity (primary)						
Incidence-Guided Saturation	0.091 (0.0008)	0.032	0.937	0.930	1.425	50
Balanced Quartiles	0.131 (0.0011)	0.017	0.936	0.827	2.138	50
Isolation Buffer	0.160 (0.0011)	0.003	0.939	0.731	3.694	21.4
High Incidence Focus	0.245 (0.0047)	0.031	0.938	0.610	4.763	50
2×2 Blocking	0.319 (0.0032)	0.010	0.939	0.642	4.244	50
Checkerboard	1.087 (0.0101)	0.001	0.941	0.547	4.737	50
Rook contiguity (sensitivity)						
Incidence-Guided Saturation	0.072 (0.0006)	0.018	0.942	0.963	1.488	50
Balanced Quartiles	0.084 (0.0006)	0.008	0.940	0.917	1.775	50
Isolation Buffer	0.133 (0.0009)	0.003	0.939	0.806	3.381	38.2
High Incidence Focus	0.206 (0.0030)	0.018	0.939	0.645	4.969	50
2×2 Blocking	0.128 (0.0012)	0.004	0.938	0.818	3.400	50
Checkerboard (flagged)	0.480 (0.0015)	−0.652	0.150	0.414	5.987	50

The Friedman test rejected equal MSE ranks across the six designs under queen contiguity at $\tau = 1$ ($\chi^2 = 451.57$, df = 5, 160 blocks, $p < 2.2 \times 10^{-16}$). Nemenyi comparisons (critical difference 0.596) separated 11 of the 15 pairs: Incidence-Guided Saturation ranked significantly better than every other design, including Balanced Quartiles ($p = 0.0086$), Balanced Quartiles ranked significantly better than the remaining four, and the four pairs not separated all lay among the bottom four designs. Holm-adjusted Wilcoxon signed-rank tests separated all 15 pairs. Because blocks within an incidence configuration share their ten surfaces, we also compared each pair of designs adjacent in the mean-MSE order over the 50 independent configuration-by-surface units; every adjacent pair differed significantly (Wilcoxon $p \leq 4.9 \times 10^{-9}$), and the better design had the lower MSE in all 50 units for every pair except High Incidence Focus against 2×2 Blocking (48 of 50). Appendix A3 gives the pairwise results and the critical-difference diagrams.

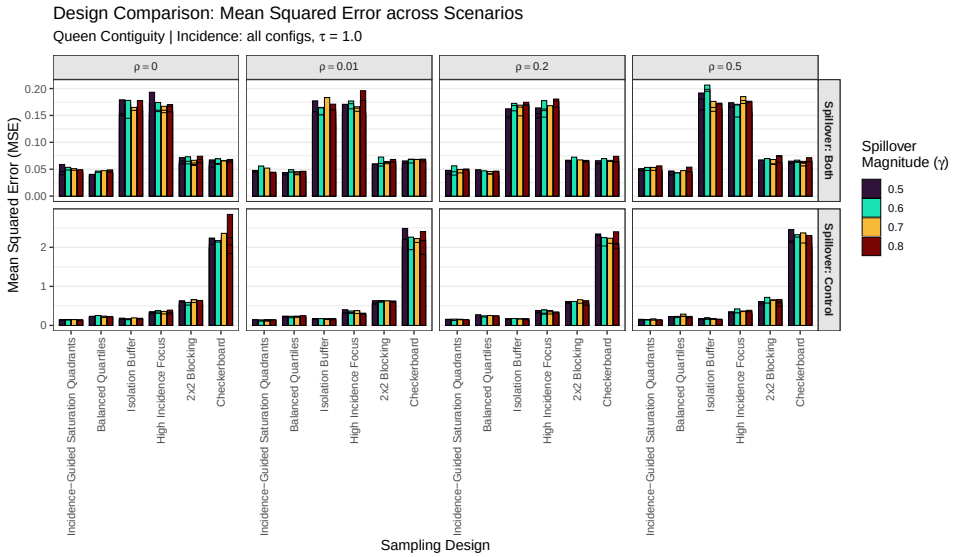


Figure 2: Mean squared error (MSE) of the oracle estimate of τ by design at $\tau = 1$, under queen contiguity. Columns of panels correspond to the four levels of outcome spatial dependence ρ, rows to the both-arms (top) and control-only (bottom) spillover regimes, and bar colors to the spillover magnitude γ. Each bar overlays the MSEs of the five incidence configurations for one combination of design, ρ, γ, and regime: the bar's height is the largest of the five, and the horizontal lines within it mark the other four. Designs are ordered by pooled mean MSE (Table 2). The vertical scale is set separately for each row. Figure 13 in Appendix A5 shows rook contiguity.

The bottom three designs were ordered differently by average rank and by mean MSE: by mean MSE, High Incidence Focus (0.245), 2×2 Blocking (0.319), and Checkerboard (1.087); by average rank, 2×2 Blocking (4.24), Checkerboard (4.74), and High Incidence Focus (4.76). Both reversals arise because the MSEs of 2×2 Blocking and Checkerboard depended strongly on the spillover regime. Each had a lower MSE than High Incidence Focus in all 80 both-arms blocks (mean MSE 0.062 for each, against 0.162) and a higher MSE in all 80 control-only blocks (0.576 and 2.111, against 0.329). Average rank counts each block once, so their wins under the both-arms regime offset their losses under the control-only regime; mean MSE weights those losses by their size, and the surface-level comparison, which averages each unit over both regimes, also separated High Incidence Focus from 2×2 Blocking.

Design rankings by incidence configuration

Table 3 gives each design's MSE within each of the five incidence configurations. Under queen contiguity, the mean-MSE order within every configuration was the same as the pooled order: Incidence-Guided Saturation, Balanced Quartiles, Isolation Buffer, High Incidence Focus, 2×2 Blocking, and Checkerboard. The configuration changed each design's MSE only slightly: in this order, each design's highest MSE across the five configurations was lower than the next design's lowest. Incidence-Guided Saturation had the lowest MSE in 18 to 21 of the 32 blocks in each configuration. Coverage did not depend on the configuration either: within each configuration, every design's coverage, averaged over its 32 scenarios, was between 0.931 and 0.943. The mechanism that generated the incidence surface therefore did not change the ranking. This includes High Incidence Focus, the design that uses the incidence ranks most directly: its MSE was no higher under the two Poisson configurations than under the iid and spatial configurations.

Table 3: Mean squared error by design within each of the five incidence configurations at $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity). Each entry averages the 32 scenarios of one design, neighbor type, and configuration (4 values of $\rho \times 4$ values of $\gamma \times 2$ spillover regimes), each summarizing 250 fits of the oracle model. The spatially autocorrelated and Poisson configurations are labeled by the strength of dependence in the incidence surface, ρ_X. Rows in both panels are ordered by pooled mean MSE under queen contiguity (Table 2). Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.

		Spatial		Poisson	
Design	iid	$\rho_X = 0.20$	$\rho_X = 0.50$	$\rho_X = 0.20$	$\rho_X = 0.50$
Queen contiguity					
Incidence-Guided Saturation	0.093	0.087	0.092	0.090	0.092
Balanced Quartiles	0.130	0.130	0.131	0.128	0.135
Isolation Buffer	0.160	0.159	0.159	0.162	0.161
High Incidence Focus	0.245	0.267	0.245	0.232	0.238
2×2 Blocking	0.311	0.315	0.323	0.326	0.320
Checkerboard	1.091	1.068	1.086	1.112	1.077
Rook contiguity					
Incidence-Guided Saturation	0.072	0.072	0.073	0.073	0.069
Balanced Quartiles	0.081	0.084	0.084	0.084	0.085
Isolation Buffer	0.131	0.131	0.132	0.136	0.134
High Incidence Focus	0.205	0.213	0.200	0.202	0.211
2×2 Blocking	0.129	0.125	0.129	0.130	0.129
Checkerboard (flagged)	0.480	0.482	0.474	0.481	0.482

Sensitivity to spatial dependence, spillover magnitude, and spillover regime

Under queen contiguity, neither the strength of spatial dependence in the outcome nor the spillover magnitude changed the ranking. Across $\rho \in \{0, 0.01, 0.20, 0.50\}$, no design's MSE changed by more than 0.02. Across $\gamma \in \{0.5, 0.6, 0.7, 0.8\}$, the MSE of every design except Checkerboard also varied by less than 0.02, and Checkerboard's varied between 1.052 and 1.108 with no trend in γ (Tables 14 and 16 in Appendix A4). Under rook contiguity, γ mattered for Checkerboard alone, for the reason given in the rook results below.

The spillover regime was the one factor that changed the queen ranking (Table 4, Figure 2). Under the both-arms regime, Balanced Quartiles had the lowest MSE (0.042), followed by Incidence-Guided Saturation (0.046), 2×2 Blocking (0.062), Checkerboard (0.062), Isolation Buffer (0.161), and High Incidence Focus (0.162). Under the control-only regime, the order was Incidence-Guided Saturation (0.135), Isolation Buffer (0.159), Balanced Quartiles (0.219), High Incidence Focus (0.329), 2×2 Blocking (0.576), and Checkerboard (2.111); at larger values of τ, Isolation Buffer matched or passed Incidence-Guided Saturation under this regime (see Stability across treatment effect sizes). For Isolation Buffer, no treated cluster has a treated neighbor, so $WZ = 0$ at every treated cluster and $WZ \circ (1 - Z) = WZ$. The two regimes in Equation 2 therefore define the same outcome model, and so the same expected MSE; the observed values, 0.161 and 0.159, differ only by Monte Carlo error, because the two regimes use separate draws. Every other design had a higher MSE under the control-only regime, and the increase was largest for Checkerboard and 2×2 Blocking, which went from close to the best designs under the both-arms regime to the two worst under the control-only regime. Among the six detailed designs, Incidence-Guided Saturation was first or second under both regimes, at this and every other value of τ, and in the full benchmark that includes SRS (Table 6) it remained competitive with SRS under both-arms spillover; this matters because a trial team may not know in advance which regime applies.

Table 4: Mean squared error by design under each spillover regime at $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages 80 scenarios (5 incidence configurations $\times$ 4 values of ρ $\times$ 4 values of γ). Rows are ordered by pooled mean MSE under queen contiguity (Table 2). Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.

	Queen		Rook	
Design	Both arms	Control only	Both arms	Control only
Incidence-Guided Saturation	0.046	0.135	0.045	0.098
Balanced Quartiles	0.042	0.219	0.042	0.125
Isolation Buffer	0.161	0.159	0.131	0.134
High Incidence Focus	0.162	0.329	0.164	0.249
2×2 Blocking	0.062	0.576	0.051	0.206
Checkerboard (flagged under rook)	0.062	2.111	0.477	0.483

The increase under the control-only regime was variance, not bias. The oracle model includes the true regime-specific spillover covariate, so it has no omitted-variable bias. At $\tau = 1$, in every combination of design and regime under queen contiguity, the absolute bias was at most 0.048 and squared bias, averaged over scenarios, was at most 2.2% of MSE. Checkerboard showed the mechanism most clearly: under the control-only regime its spillover covariate is a linear combination of the intercept and Z at every interior cluster, so only the boundary clusters separate the spillover effect from the treatment effect (Methods). Checkerboard's MSE rose from 0.062 under the both-arms regime to 2.111 under the control-only regime, while its bias stayed near zero (0.004) and its coverage stayed at 0.942: its intervals widened with its variance. Its pooled MSE of 1.087 was almost entirely variance: squared bias, averaged over scenarios, was 0.0045, less than 0.5% of MSE (Figure 12 in Appendix A5 shows the decomposition for the Poisson configuration with $\rho_X = 0.20$). 2×2 Blocking showed the same pattern on a smaller scale, with MSE rising from 0.062 to 0.576 and a bias of 0.016 under the control-only regime.

Rook contiguity

Under rook contiguity at $\tau = 1$, pooled over the five incidence configurations, the mean-MSE order was Incidence-Guided Saturation (0.072), Balanced Quartiles (0.084), 2×2 Blocking (0.128), Isolation Buffer (0.133), High Incidence Focus (0.206), and Checkerboard (0.480) (Table 2; Figure

13 in Appendix A5), and the same order held within each of the five incidence configurations (Table 3). Incidence-Guided Saturation had the lowest MSE in 97 of the 160 rook blocks (61%) and Balanced Quartiles in 60 (38%). Coverage was between 0.938 and 0.942 for every design except Checkerboard, whose coverage was 0.150 (Figure 14 in Appendix A5). The Friedman test rejected equal ranks ($\chi^2 = 703.76$, df $= 5$, 160 blocks, $p < 2.2 \times 10^{-16}$). Holm-adjusted Wilcoxon tests separated every pair except 2×2 Blocking against Isolation Buffer, which were close and traded places at some values of τ; Nemenyi comparisons also left Incidence-Guided Saturation and Balanced Quartiles unseparated, but Incidence-Guided Saturation had the lower MSE in 48 of the 50 independent configuration-by-surface units (Appendix A3).

Checkerboard under rook contiguity is a different case: τ is not identified, and the coefficient on Z estimates $\tau - \gamma$ (Methods). At $\tau = 1$, Checkerboard's mean bias was -0.502, -0.600, -0.704, and -0.802 at $\gamma = 0.5$, 0.6, 0.7, and 0.8, while its mean standard deviation stayed between 0.20 and 0.21; squared bias, averaged over scenarios, accounted for 0.438 of its MSE of 0.480 (Figure 15 in Appendix A5). Its coverage fell from 0.316 at $\gamma = 0.5$ to 0.027 at $\gamma = 0.8$, and was between 0.134 and 0.161 at every level of ρ, against 0.939 to 0.943 under queen contiguity (Table 15 in Appendix A4). Because $WZ = 1 - Z$, the two spillover regimes define the same outcome model; the observed MSEs, 0.477 under the both-arms regime and 0.483 under the control-only regime, differ only by Monte Carlo error. Rook Checkerboard estimates were precise but centered on the wrong quantity, the opposite of its imprecise but valid queen estimates. We keep these estimates in every ranking and flag them.

Non-oracle sensitivity analysis

The non-oracle model, which omits the spillover term, was biased for every design (Table 17 in Appendix A6 gives both models' MSE, bias, and coverage for each design). At $\tau = 1$, and setting aside Checkerboard under rook contiguity, its bias ranged from -0.12 to -0.40 and its coverage from 0.51 to 0.90 across designs and neighbor types; for Checkerboard under rook contiguity the

two models coincide, because once the aliased spillover column is dropped they are the same. Omitting the spillover term also removes a column that can be nearly collinear with Z, and the non-oracle estimates were accordingly less variable for some designs. Under queen contiguity, the non-oracle MSE was lower than the oracle MSE for every design, most of all for Checkerboard (oracle 1.087, non-oracle 0.108) and 2×2 Blocking (oracle 0.319, non-oracle 0.121); under rook contiguity it was higher for every design except 2×2 Blocking (0.128 under both models) and Checkerboard (identical). Where a design's non-oracle MSE was lower, however, its estimate was biased and its intervals under-covered, so the lower MSE does not support valid inference. These results do not change the recommendation: Incidence-Guided Saturation also had the lowest non-oracle MSE under both neighbor types (0.087 under queen contiguity and 0.086 under rook contiguity).

Stability across treatment effect sizes

Under queen contiguity, each of the two design orders was the same at every $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$: the order by average within-block rank did not change across these values, and neither did the order by mean MSE, although the two orders differ from each other in the bottom three designs (see Design performance under queen contiguity; Figure 3a shows mean MSE). The Friedman statistic was 455.49, 451.57, 429.01, 411.19, and 376.79 at these five values ($p < 2.2 \times 10^{-16}$ at each). Under rook contiguity the order was also stable (Figure 3b) except for the close pair of 2×2 Blocking and Isolation Buffer: by average rank, 2×2 Blocking was ahead only at $\tau = 0.8$, and by mean MSE, Isolation Buffer was ahead only at $\tau = 3$. The five values of τ share their assignments and noise, so these five comparisons are not independent confirmations of one another; they show that, under queen contiguity, the order did not change across the five values of τ we tested, given the same draws.

Within that stable order, the margins moved. Under queen contiguity, the gap in MSE between Incidence-Guided Saturation and Balanced Quartiles stayed between 0.037 and 0.041, and

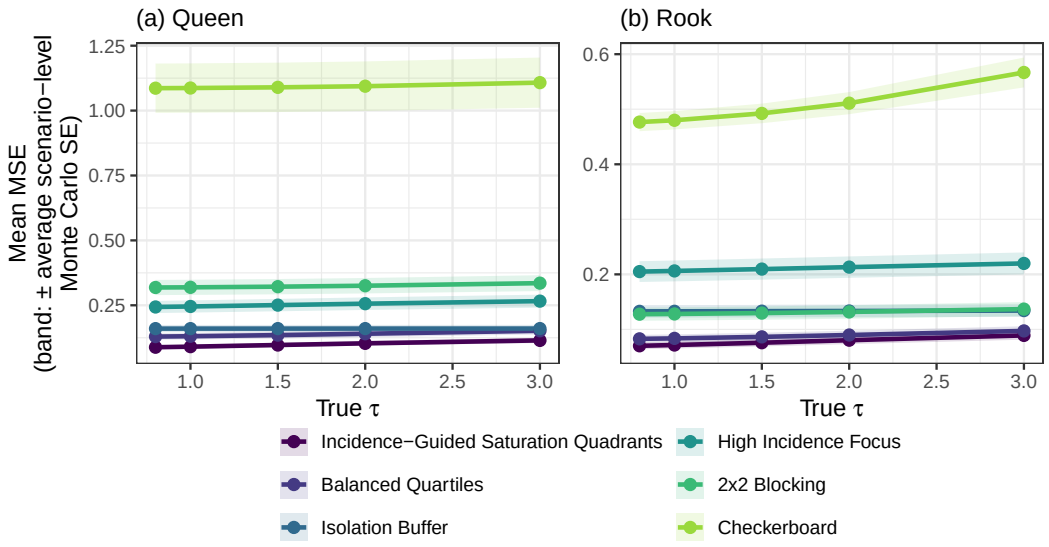


Figure 3: Mean squared error by design across the five values of the treatment effect, $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$, under (a) queen contiguity (primary) and (b) rook contiguity (sensitivity). Each point averages the 160 scenarios of one design at one value of τ, pooled over the five incidence configurations. Shaded bands show ± 1 times the average of the scenario-level Monte Carlo standard errors of MSE; they describe the precision of a single scenario's MSE, not of the plotted average, whose standard error is smaller (Table 2). The shared legend lists the designs in order of pooled mean MSE under queen contiguity at $\tau = 1$. The vertical scales of the two panels differ. The five values of τ share their assignments and noise.

Incidence-Guided Saturation had the lower MSE of the two in 101 to 103 of the 160 blocks at every value of τ. Isolation Buffer closed the gap instead: its MSE stayed at 0.160 at every value of τ, while that of Incidence-Guided Saturation rose from 0.088 to 0.115 and that of Balanced Quartiles from 0.130 to 0.152. Isolation Buffer's gain came under the control-only regime, where the mean MSE of Incidence-Guided Saturation rose from 0.131 at $\tau = 0.8$ to 0.182 at $\tau = 3$, while that of Isolation Buffer stayed between 0.159 and 0.160. At $\tau = 2$ the two were level (0.1596 for Isolation Buffer against 0.1604), and at $\tau = 3$ Isolation Buffer had the lower control-only MSE (0.160 against 0.182). At $\tau = 2$ and $\tau = 3$, Isolation Buffer also had the lower worst case across all 160 blocks (0.207 against 0.210, and 0.208 against 0.238). Incidence-Guided Saturation remained second under the control-only regime at those two values, second to Balanced Quartiles under the both-arms regime at every value of τ, and lowest in pooled MSE at every value of τ. The rise in its MSE was mostly variance: its mean bias grew

from 0.029 at $\tau = 0.8$ to 0.047 at $\tau = 3$, but squared bias stayed at or below 3.0% of its MSE. Appendix A7 traces both rises to a term of the estimation error that grows with τ when treated clusters have treated neighbors, and gives the block-level detail.

Under rook contiguity, Incidence-Guided Saturation kept the lowest control-only MSE at every value of τ, although its margin over Isolation Buffer narrowed, from 0.095 against 0.134 at $\tau = 0.8$ to 0.131 against 0.136 at $\tau = 3$.

Except for Checkerboard under rook contiguity, coverage stayed close to 0.94 at every value of τ: between 0.936 and 0.941 for every design under queen contiguity, and between 0.937 and 0.942 for every other design under rook contiguity. The coverage of rook Checkerboard, for which τ is not identified, rose from 0.138 at $\tau = 0.8$ to 0.442 at $\tau = 3$, because its intervals widened around the wrong quantity (Appendix A7). Appendix A3 shows coverage across τ for each design.

Robustness across blocks

Table 5 gives the distribution of MSE across the 160 queen blocks at $\tau = 1$. Incidence-Guided Saturation had the lowest median (0.080) and the lowest worst-case MSE of any design (0.169). Isolation Buffer had the narrowest range (0.128 to 0.206), because its MSE does not depend on the spillover regime, but at a higher level. Balanced Quartiles had a low median (0.115) and a worst case of 0.290. The distributions of 2×2 Blocking and Checkerboard reflected their dependence on the spillover regime: both had low first quartiles (0.062 and 0.063) and high upper quartiles (0.580 and 2.093). Checkerboard had the highest MSE in all 80 control-only blocks; in the 80 both-arms blocks, the highest MSE belonged to High Incidence Focus in 42 and to Isolation Buffer in 38. Under rook contiguity, Incidence-Guided Saturation also had the lowest worst-case MSE (0.138). These worst cases summarize variation across parameter settings; they are distinct from the conditional risk across assignments within one setting, examined next.

Table 5: Distribution of block-level mean squared error by design across the 160 blocks under queen contiguity at $\tau = 1$, pooled over the five incidence configurations (32 blocks in each). At a fixed value of τ, each block contributes one scenario per design. Rows are sorted by median, which here coincides with the order of pooled mean MSE under queen contiguity (Table 2).

Design	Best	Q25	Median	Q75	Worst
Incidence-Guided Saturation	0.037	0.045	0.080	0.135	0.169
Balanced Quartiles	0.034	0.042	0.115	0.221	0.290
Isolation Buffer	0.128	0.150	0.159	0.169	0.206
High Incidence Focus	0.139	0.160	0.234	0.322	0.424
2×2 Blocking	0.048	0.062	0.271	0.580	0.719
Checkerboard	0.051	0.063	0.899	2.093	2.843

SRS benchmark and allocation-specific risk

At $\tau = 1$ under queen contiguity, pooled over the five incidence configurations and all spatial/spillover settings, Incidence-Guided Saturation has mean MSE 0.0907 against SRS 0.1285; Saturation Quadrants has 0.0900. Table 6 separates the regimes and includes all eight strategies, and Figure 4 shows the six detailed designs and SRS with the five incidence configurations kept separate: within each regime the configurations lie close together for every design, whereas the regime changes how each design compares with SRS. Incidence-Guided Saturation's control-only ratio to SRS is 0.631, whereas its both-arms ratio is 1.083. SRS, Balanced Quartiles and Balanced Halves have similar both-arms accuracy. Small mean differences should not be interpreted as equivalence, nor should pooling the regimes obscure the difference in design guidance. Paired comparisons over 50 independent configuration-by-surface units support the lower pooled MSE of both saturation designs relative to SRS; the six-design Friedman rankings elsewhere describe that subset only.

To assess allocation risk, we additionally conducted a focused grid pilot with queen weights, $\tau = 1$, two fixed surfaces from each incidence configuration and the extreme combinations of ρ and γ, $\rho \in \{0, 0.5\}$ and $\gamma \in \{0.5, 0.8\}$. For assignment a, its risk is the conditional MSE $m(a) = E_\epsilon[(\hat{\tau} - \tau)^2 \mid a, X]$. We estimated it by simulating repeated outcomes for each sampled assignment and then compared its distribution across assignments. This is different from the distribution of squared errors in a single allocation or MSE across parameter scenarios. The

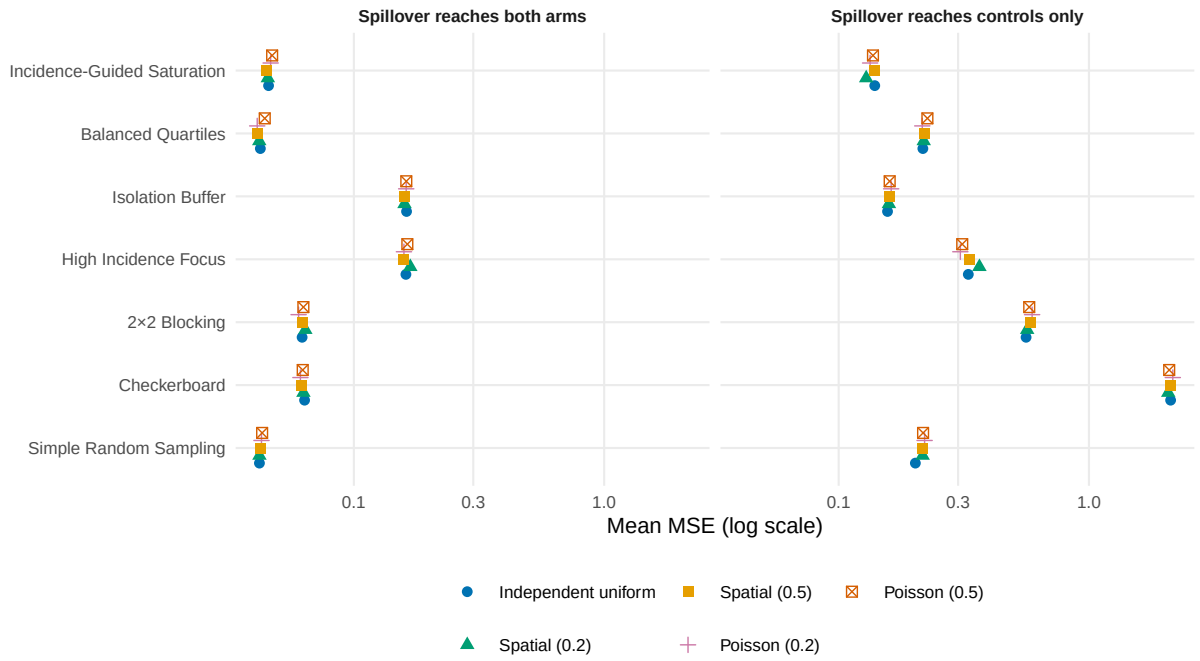


Figure 4: Grid mean MSE at $\tau = 1$ with queen weights, for the six designs and simple random sampling (SRS), under each spillover regime. Points keep the five incidence configurations separate, each averaging the 16 combinations of outcome spatial dependence ρ and spillover magnitude γ within that configuration and regime; parentheses in the legend give incidence dependence ρ_X. Both panels use the same logarithmic scale. Table 6 gives the values pooled over configurations for all nine strategies.

Table 6: Full grid benchmark at $\tau = 1$, queen contiguity. Each regime-specific entry averages 80 scenarios (five incidence configurations, four ρ and four γ values). MSE is the mean squared error of the estimate of τ; coverage is the proportion of nominal 95% confidence intervals that contain τ. Both use the oracle fit, the spatial lag model that includes the true spillover term. Simple random sampling is complete randomization of 50 clusters. No value pools queen and rook.

	Both arms		Control only	
Design	MSE	Coverage	MSE	Coverage
Checkerboard	0.0624	0.940	2.1114	0.942
High Incidence Focus	0.1617	0.941	0.3292	0.935
Saturation Quadrants	0.0442	0.942	0.1359	0.929
Isolation Buffer	0.1608	0.939	0.1595	0.939
2×2 Blocking	0.0623	0.937	0.5761	0.941
Balanced Quartiles	0.0421	0.940	0.2195	0.932
Balanced Halves	0.0420	0.939	0.2179	0.933
Incidence-Guided Saturation	0.0459	0.939	0.1355	0.934
Simple Random Sampling	0.0424	0.941	0.2146	0.936

pilot used 100 assignment draws for each randomized design and 100 simulated outcomes per distinct assignment, and selected settings were refined to 400 outcomes on the same assignments; a design with only one possible assignment received its own outcome replication.

Averaged over the 40 units per regime in the pilot (fixed surfaces crossed with the extreme combinations of ρ and γ), Incidence-Guided Saturation's control-only mean MSE is 0.1409 against SRS 0.2172, the estimated 90th percentile of allocation-specific MSE is 0.1747 against 0.2980, and the estimated worst-decile mean is 0.1936 against 0.3486. Under both arms, the corresponding mean MSE is 0.0458 against 0.0428 and worst-decile mean 0.0582 against 0.0544. Saturation Quadrants also performs favorably under control-only spillover; Table 9 in the application section compares every design with SRS on both criteria. These selected-unit results support a qualified downside argument for saturation; they are not the full-study grid estimates. The focused refinement preserves the control-only direction but reveals substantial outcome-noise contributions to estimated allocation variation, especially under both arms. Quantiles and worst-decile membership estimated from a finite number of outcomes remain uncertain. We therefore use the completed NC application below as the next setting-specific assessment rather than extrapolate exact tail protection from this pilot.

Application: Observed NC SUD Incidence for Education-Pilot Planning

Purpose and observed inputs

The application asks which allocation designs estimate a known education-intervention effect accurately on this geography, when some designs use historical SUD incidence to direct treatment. It supports planning an education intervention for healthcare professionals involved in SUD prevention. There are no observed intervention outcomes and no estimated relationship between local SUD incidence and education ability. Any eventual effect on SUD incidence requires a separate clinical evaluation. A real pilot might realize only one allocation per planning year, making both average estimation error and poor-allocation downside relevant.

The assignment units are 58 community college service-area clusters that partition North

Carolina's 100 counties under the study mapping (Figure 5). Northampton is assigned to Halifax Community College, Bertie to Martin Community College, and Hertford alone to Roanoke-Chowan Community College; these are analysis conventions, not exclusive service commitments. Observed county-year deaths and population estimates for ages 18–64 from the Surveillance, Epidemiology, and End Results program were aggregated to the same named clusters for each year, 2018–2021. The corrected numerator supplied by Habib includes heart-failure deaths formerly excluded as presumed non-sudden (A. Habib, personal communication, September 25, 2026). It totals 23,523 deaths over 25,594,321 person-years. The earlier numerator reproduces the totals behind Habib's unpublished report [29]; the corrected numerator is the application input. We do not infer undocumented upstream exclusion rules or zip-code allocation from the received county panel.

Annual incidence is $100,000 D_{iy}/P_{iy}$, and pooled incidence is $100,000 \sum_y D_{iy} / \sum_y P_{iy}$, with the latter population sum interpreted as person-years. All 58 clusters are present in each year, and counts and populations sum back to statewide totals. We hold these observed annual surfaces fixed, generate no synthetic incidence surface and report years separately (Figure 6). The county death-certificate source data are restricted and not publicly available; only 58-cluster summaries are reported.

Geographic adaptations and treatment budgets

Queen contiguity between service areas, computed from legal county boundaries, defines primary adjacency, and rook contiguity is a sensitivity analysis. Shared legal boundaries across water can create adjacency; this is a prespecified geographic exposure proxy, not a measured professional communication network.

All eight designs and SRS are evaluated under the names used on the grid, adapted to the irregular map; Appendix A8 gives the rules in full and maps the fixed regions and example assignments. Checkerboard becomes a deterministic greedy graph-coloring assignment, an

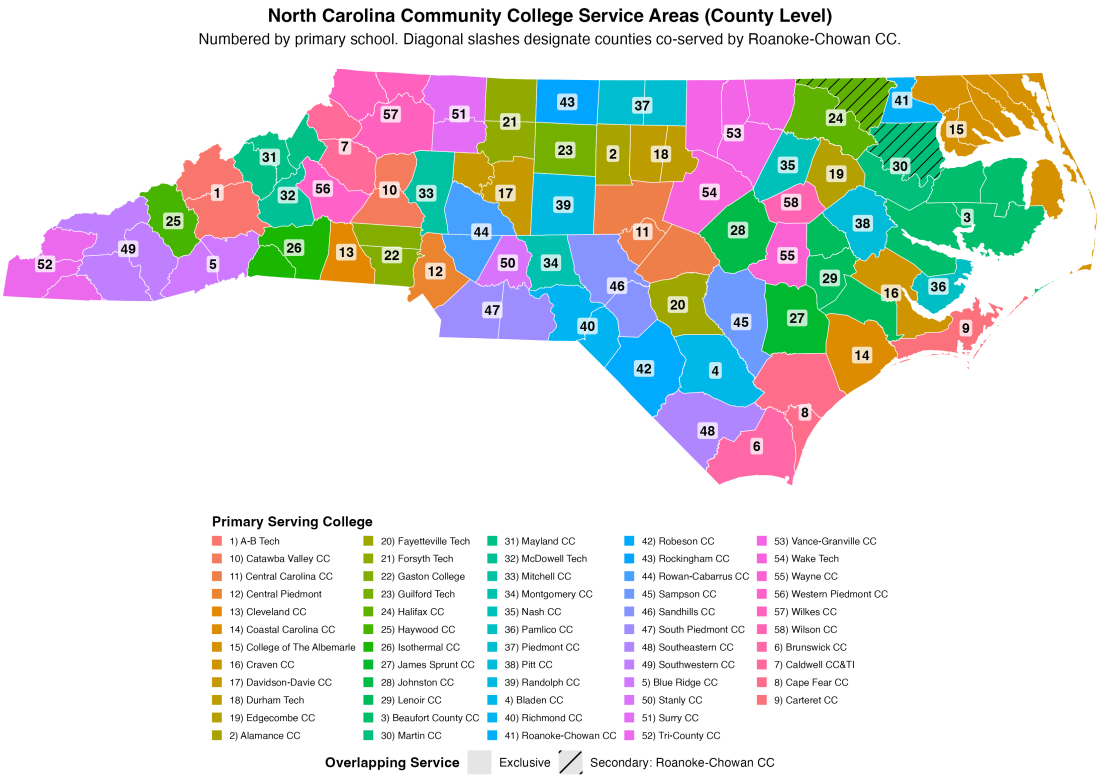


Figure 5: The assignment of North Carolina's 100 counties to the 58 community college service areas used as clusters in the planned application. Each county is colored by its primary serving college, and each service area carries a single number, keyed to the legend. Hatching marks the two counties that also lie within Roanoke-Chowan Community College's broader service network: Northampton County, assigned to Halifax Community College (24), and Bertie County, assigned to Martin Community College (30). Roanoke-Chowan Community College (41) serves Hertford County alone in this analysis. The map shows county boundaries and the county-to-college assignment only; it does not depend on the SUD data.

interspersion analog rather than a literal rectangular checkerboard or an exact bipartite coloring. High Incidence Focus treats the 29 highest annual-rate clusters, Isolation Buffer draws a greedy random maximal independent set on the NC adjacency graph, SRS treats a random 29 of 58 clusters, and Balanced Quartiles and Balanced Halves stratify annual rate ranks and treat 29 overall. The two saturation designs use one fixed four-region partition for all years, chosen for geographic compactness and balance without consulting outcomes. Saturation Quadrants randomly assigns 20/40/60/80% to these regions each draw; Incidence-Guided Saturation ranks the regions by the mean rank of their clusters' annual per-capita rates and assigns 80/60/40/20% from highest to lowest; mean raw rates and pooled regional rates serve as prespecified alternative regional summaries. On the NC map, 2×2 Blocking uses spatial blocks of nearby clusters,

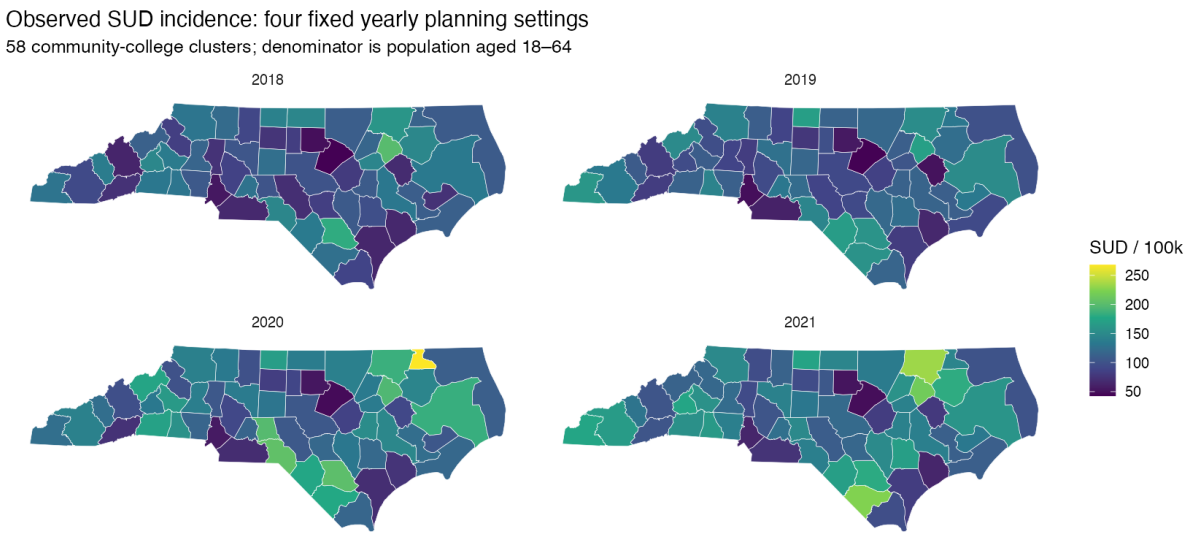


Figure 6: Observed annual SUD incidence per 100,000 adults aged 18–64 in the 58 service-area clusters. These four fixed planning inputs direct incidence-aware allocation; they are not simulated outcomes or measures of education ability.

503 14 irregular groups formed once per year, and treats half of each group, rounded, at random.

504 Budgets therefore differ across designs; we report them rather than modify the designs to force

505 equal budgets (Table 7).

Table 7: Primary queen application treatment budgets. Entries are mean treated-cluster counts / percent of annual population aged 18–64. Counts and population shares use the same sampled assignments; population shares are recalculated for each year, including for designs whose assignments do not depend on annual incidence. Shares are geographic reach, not healthcare-professional enrollment.

Design	2018	2019	2020	2021
Checkerboard	27.00 / 50.4	27.00 / 50.4	27.00 / 50.4	27.00 / 50.7
High Incidence Focus	29.00 / 22.2	29.00 / 23.5	29.00 / 23.3	29.00 / 22.4
Saturation Quadrants	29.03 / 49.2	29.03 / 49.2	29.03 / 49.1	29.03 / 49.1
Isolation Buffer	15.89 / 26.7	15.89 / 26.7	15.89 / 26.7	15.89 / 26.7
2×2 Blocking	28.00 / 50.0	31.00 / 55.1	27.00 / 48.0	29.00 / 50.3
Balanced Quartiles	29.00 / 48.5	29.00 / 49.7	29.00 / 50.6	29.00 / 49.8
Balanced Halves	29.00 / 50.0	29.00 / 50.5	29.00 / 49.2	29.00 / 49.4
Incidence-Guided Saturation	29.00 / 53.2	29.00 / 45.1	29.00 / 47.9	29.00 / 51.8
Simple Random Sampling	29.00 / 49.5	29.00 / 49.5	29.00 / 49.5	29.00 / 49.5

Outcome model and Monte Carlo evaluation

For the primary education outcome, we set $\beta = 0$ in Equation 1, use the 58-cluster W, and fix $\tau = 1$, $\sigma = 1$:

$$Y = (I - \rho W)^{-1}[\tau Z + S(Z) + \epsilon], \qquad \epsilon \sim N(0, I). \tag{3}$$

Observed incidence affects assignments but neither baseline education nor the constant structural intervention effect. We estimate τ with the same validated maximum likelihood implementation as in the grid study, with an intercept, Z and the true regime-specific $S(Z)$, without X. As in the grid model, τ is the coefficient before spatial propagation, not an average total impact after applying $(I - \rho W)^{-1}$.

Primary queen settings cross $\rho = 0, 0.2, 0.5$, $\gamma = 0.5, 0.8$, both regimes, four years and nine designs (432 reported settings). A matched hypothetical baseline sensitivity adds $X = (\text{rank}_{\text{average}}(rate) - 0.5)/58$ with $\beta = 1$ to both the outcome baseline and the fitted covariates (432 settings). It changes baseline and adjustment together and is not an empirical incidence–education association. A rook-contiguity sensitivity analysis of the primary model uses the extreme values $\rho = 0, 0.5$ (288 settings), and the two alternative regional summaries add 96 Incidence-Guided Saturation settings. The full application reports 1,248 settings, computed from 936 distinct simulation settings, because reported settings with the same allocations and outcome model share one simulation.

We sampled $J = 100$ assignments and simulated $R = 100$ independent outcomes per distinct assignment; a design whose set of possible assignments contains only one assignment instead received $R = 1,000$ outcomes for that assignment, rather than 100 repeated copies of it counted as separate allocations. A prespecified adaptive schedule extended allocations or outcomes where needed until the Monte Carlo standard error of mean MSE was at most 5% of mean MSE and that of coverage at most 0.01; two single-assignment settings in the matched sensitivity analysis required $R = 2,000$, and all randomized designs met these targets at $J = 100$ and

$R = 100$ (Appendix A8). In the primary model, Checkerboard, Saturation Quadrants, Isolation Buffer and SRS reuse the same simulations across years, because annual incidence enters neither their allocation nor the outcome model. These repeated values are not independent annual confirmation.

For each allocation a, the repeated outcomes estimate the conditional MSE $m(a) = E_\epsilon[(\hat{\tau} - \tau)^2 \mid a, \text{observed inputs}]$, the expected squared error of the estimate if a were the allocation realized. We report its distribution across sampled allocations, its estimated 90th percentile and the estimated mean among the worst 10% of sampled allocations, alongside unconditional mean MSE, bias and coverage. A separate tail refinement at queen $\rho = 0, 0.5$, $\gamma = 0.8$, both regimes and all years and designs kept the same allocations and extended the randomized designs to $R = 400$ outcomes per allocation, keeping $R = 1,000$ for single-assignment designs; it is not a new allocation sample. Worst-decile assignments were identified with one half of the simulated outcomes and evaluated with the other half, to avoid selection bias. Design-matrix rank, residual treatment information, aliased columns, warnings, invalid estimates or intervals, and boundary estimates of ρ were recorded.

Completed results: accuracy, inference and reach

The main run generated 7,598,000 independent outcome fits. Every setting is complete and meets its prespecified precision targets for mean MSE and coverage: the largest relative Monte Carlo standard error of mean MSE is 0.04871 and the largest Monte Carlo standard error of coverage is 0.00892. There are no warnings, aliased columns, estimates of ρ at the boundary of its parameter space or estimation failures. The tail refinement adds 2,280,000 independent outcomes; counting the outcomes reused from the main run only once gives 9,878,000 distinct outcomes across the two stages. Monte Carlo precision describes uncertainty in simulated performance; it does not certify interval calibration.

Figure 7 compares the eight designs with SRS by year and regime. Under control-only spillover,

Saturation Quadrants has descriptive mean MSE 0.22329 against SRS 0.30761, a 27.4% reduction. Incidence-Guided Saturation reduces mean MSE by 21.1–29.9% across the four observed years (Table 8). Both saturation designs have lower mean MSE than SRS in every primary parameter setting, and approximate Monte Carlo intervals for each setting-specific difference excluded zero. Neither is uniquely best: as on the grid, the two performed similarly, and their advantage over SRS comes from the saturation structure both designs share. High Incidence Focus is year-sensitive under control-only spillover and cannot be recommended uniformly.

Under both-arms spillover, SRS mean MSE is 0.07638, Saturation Quadrants 0.07518, and Incidence-Guided Saturation ranges from 0.07428 to 0.07682 across years. SRS remains competitive with balancing, blocking and saturation. Small point differences do not establish equivalence or universal optimality. High Incidence Focus has low both-arms MSE in the primary model, but its allocation is deterministic for these fixed rates; this is conditional model-based accuracy, not evidence of randomization-based causal validity.

Mean annual coverage across primary settings is approximately 92.6–93.6%, below nominal 95%. The full year/design/regime bias and coverage table is in Appendix A8. Meeting the Monte Carlo precision targets does not repair this finite-sample undercoverage. Checkerboard's treatment effect is identified on this irregular graph, but weak treatment–spillover separation makes its estimates imprecise; this differs from exact rook non-identification on the rectangular grid.

Population shares contextualize performance (Table 7). High Incidence Focus treats 29 clusters but only about 22–24% of working-age population, while Isolation Buffer treats a mean of 15.89 clusters and about 27%. These population denominators measure geographic reach, not the number of healthcare professionals educated, intervention cost or participant-level precision.

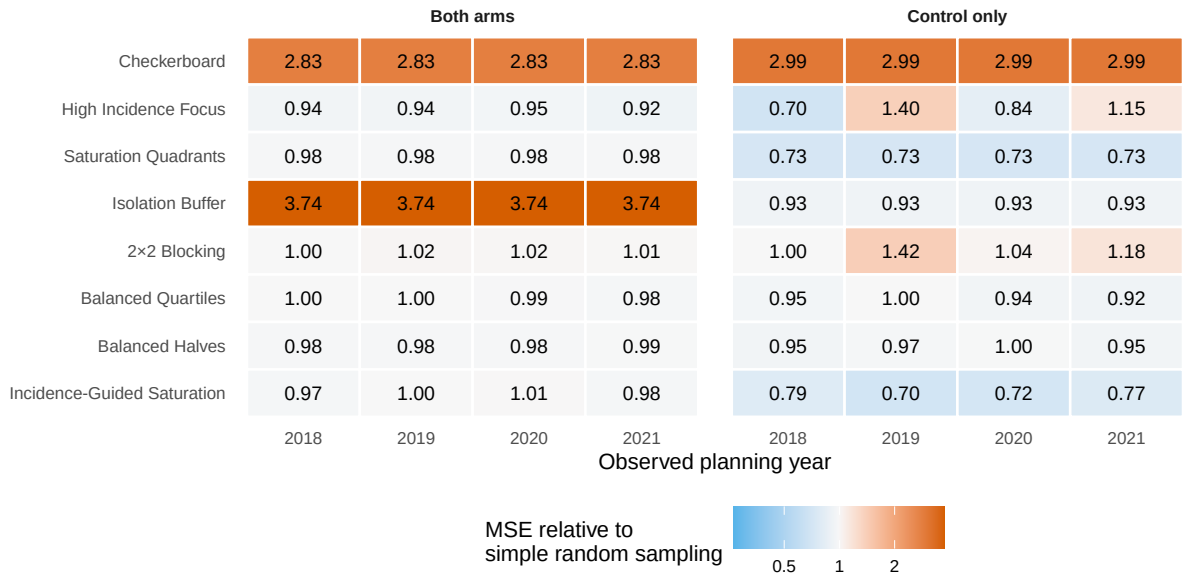


Figure 7: Completed primary application: yearly mean MSE divided by year- and regime-matched SRS MSE, for the eight other designs with queen weights and education outcomes. Each cell averages six ρ/γ settings equally before taking the ratio. Blue indicates lower error than SRS, orange higher error; colors use a logarithmic ratio scale and cells print the ratios. SRS mean MSE is 0.07638 under both arms and 0.30761 control-only. Years that share the same simulations are not independent replications. Absolute MSE, bias, coverage, budgets and setting-specific comparisons are in Appendix A8.

Table 8: Primary application yearly mean MSE, queen contiguity; equal weights over the six ρ/γ pairs. Saturation Quadrants and SRS reuse the same simulations across years.

	Both arms		Control only		
Year	SRS	Incidence-Guided Saturation	SRS	Saturation Quadrants	Incidence-Guided Saturation
2018	0.07638	0.07428	0.30761	0.22329	0.24285
2019	0.07638	0.07628	0.30761	0.22329	0.21565
2020	0.07638	0.07682	0.30761	0.22329	0.22029
2021	0.07638	0.07450	0.30761	0.22329	0.23732

Allocation downside and sensitivity

At the two refined queen settings, $\rho = 0$ and $\rho = 0.5$ with $\gamma = 0.8$, under control-only spillover, Saturation Quadrants's estimated 90th percentile of allocation-specific MSE is 31.8–32.9% below SRS and its estimated worst-decile mean 34.6–37.8% below. Incidence-Guided Saturation's corresponding reductions are 20.8–38.6% and 20.7–43.4%. The pattern persists under the split-half evaluation described above. Figure 19 in Appendix A8 shows the $\rho = 0.5$ setting. This supports a descriptive downside benefit alongside lower average error, conditional on the studied

model and sampled assignments.

Table 9 compares every design with SRS on both criteria, mean conditional MSE and the mean of the worst 10% of sampled allocations, in the grid pilot and at the refined NC settings. Under control-only spillover, both saturation designs had lower mean and worst-decile MSE than SRS in both settings. Isolation Buffer also had a lower worst decile than SRS under control-only spillover (0.62 of SRS in the grid pilot and 0.91 in NC), but under both-arms spillover its mean was 3.79 and 3.69 times that of SRS, and it treated fewer clusters. High Incidence Focus has a single allocation in NC. That allocation's MSE was below the SRS worst-decile mean under both regimes (0.76 under both arms and 0.64 under control-only spillover), with mean MSE close to SRS (0.93 and 1.03). In the grid pilot, however, its both-arms mean was 3.91 times that of SRS, and its NC result depends on the incidence model (next paragraph). Balanced Quartiles and Balanced Halves matched SRS on both criteria. Checkerboard and 2×2 Blocking did not improve on SRS on either criterion in any setting.

Table 9: Mean conditional MSE and worst-decile mean conditional MSE of each design relative to SRS: each entry is the design's average divided by SRS's average in the same setting and spillover regime (below 1 is lower error than SRS). Grid: focused pilot, queen contiguity, $\tau = 1$, averaged over 40 units, each a fixed incidence surface with one of the extreme combinations of ρ and γ. NC: refined queen settings $\rho \in \{0, 0.5\}$, $\gamma = 0.8$, averaged with equal weight over 2018–2021 and both values of ρ; NC designs are the map adaptations (Checkerboard as a greedy graph-coloring assignment, 2×2 Blocking on spatial blocks of nearby clusters, and both saturation designs on four fixed regions). A design with one allocation has worst-decile mean equal to its mean.

	Grid pilot				NC application			
	Both arms		Control only		Both arms		Control only	
Design	Mean	Worst 10%	Mean	Worst 10%	Mean	Worst 10%	Mean	Worst 10%
Checkerboard	1.41	1.10	9.80	6.11	2.68	2.20	3.02	1.88
High Incidence Focus	3.91	3.11	1.38	0.87	0.93	0.76	1.03	0.64
Saturation Quadrants	1.06	1.05	0.62	0.53	0.99	0.94	0.73	0.64
Isolation Buffer	3.79	3.93	0.75	0.62	3.69	4.75	0.93	0.91
2×2 Blocking	1.44	1.57	2.68	2.63	1.00	1.02	1.15	1.18
Balanced Quartiles	0.99	1.00	1.00	1.00	0.98	0.96	0.95	1.01
Balanced Halves	0.99	1.00	1.00	0.98	0.98	0.96	0.97	0.99
Incidence-Guided Saturation	1.07	1.07	0.65	0.56	0.98	0.93	0.75	0.68
Simple Random Sampling	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Figure 8 shows the full allocation-to-allocation distribution for the six designs and SRS. Under control-only spillover, most Incidence-Guided Saturation allocations had lower MSE than the

average SRS allocation in both studies, whereas the SRS distribution extended well above its mean.

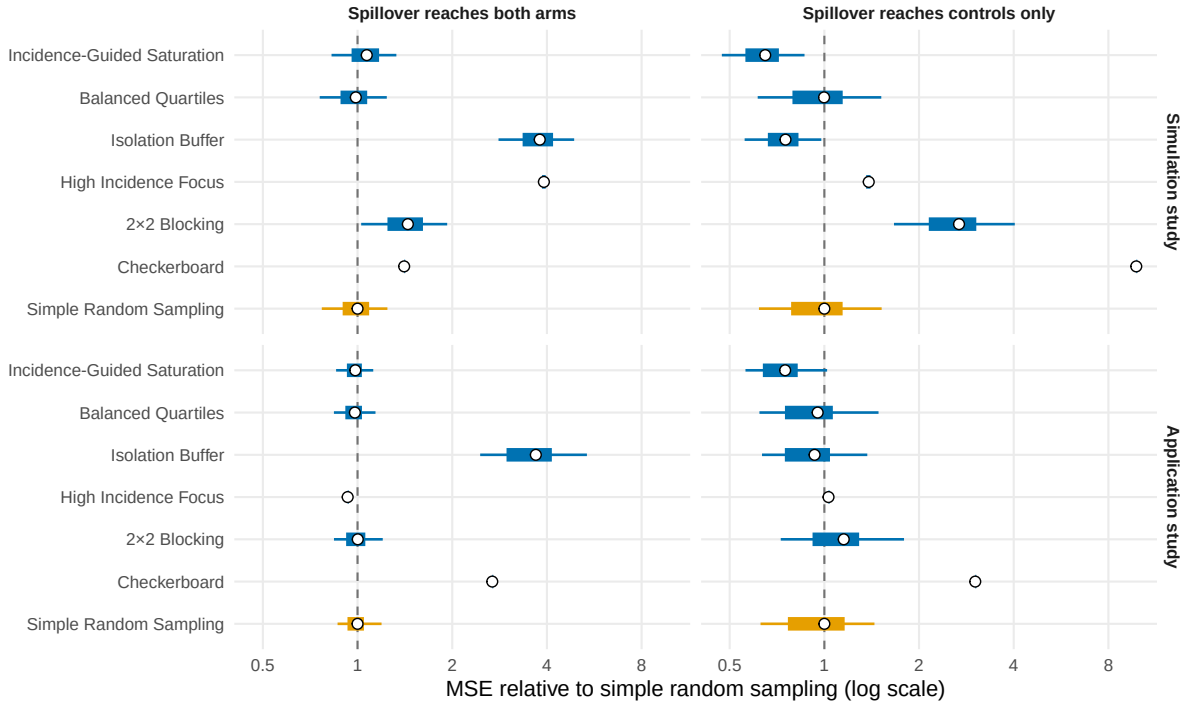


Figure 8: Distribution of allocation-specific MSE for the six designs and simple random sampling (SRS), relative to the mean MSE of SRS in the same setting (queen weights). Simulation study: focused grid pilot at $\tau = 1$, 40 fixed surface/parameter settings per spillover regime. Application study: refined NC settings ($\rho = 0, 0.5$; $\gamma = 0.8$) in each year 2018–2021. For each allocation, MSE is estimated from repeated simulated outcomes (100 in the simulation study, at least 400 in the application study). Thin lines span the 5th to 95th percentiles of allocations and thick bars the 25th to 75th; circles mark the mean. Percentiles are computed within each setting and then averaged, so bars show allocation-to-allocation spread; a design with one allocation per setting appears as a circle only. Outcome Monte Carlo noise in the per-allocation estimates widens the bars slightly.

Tail uncertainty remains material. With $J = 100$, the worst decile contains only ten sampled draws; extra outcomes reduce noise in $\hat{m}(a)$ but do not enlarge the allocation sample. All 112 reported settings in the tail refinement for designs with more than one possible assignment flag uncertain worst-decile membership. Under both arms, the overlap between the worst deciles identified by the two outcome halves (the Jaccard index: allocations in both divided by allocations in either) was 0–0.25 for Incidence-Guided Saturation, and approximately 61–86% of the raw variance of estimated risk across allocations was attributable to outcome Monte Carlo noise. Saturation Quadrants's worst-decile membership was similarly unstable. Small both-arms

tail differences therefore do not establish superior downside protection. Checkerboard and High Incidence Focus each have only one possible assignment for these inputs; their 90th percentile and worst-decile mean equal their conditional mean MSE, which does not imply low error or absence of outcome uncertainty.

In the matched hypothetical incidence-baseline/X-adjusted sensitivity, High Incidence Focus MSE rises to roughly 3.68–4.44 times its primary value under both arms and 1.42–2.52 times under control-only spillover. Treatment–incidence alignment leaves less residual treatment information after adjustment; this sensitivity changes baseline and adjustment jointly and does not imply that SUD incidence predicts education ability. Rook and queen comparisons matched at the same extreme combinations of ρ and γ preserve broad regime-specific conclusions. Alternative regional summaries change the regional ordering in some years; where an alternative summary gives the same ordering and ties, it gives the same set of possible assignments and so the same results. We retain the prespecified mean-rank rule rather than select a summary retrospectively. Appendix A8 reports the matched sensitivity table and regional ordering.

The application supports considering either saturation strategy when control-only spillover is plausible, and SRS alongside other randomized strategies when both-arm exposure is expected. A trial team should consider likely spillover, recruitment, cluster and participant budgets, achievable measurement precision and interval calibration before selecting one realized allocation. The findings do not establish intervention effectiveness or a final trial protocol.

Discussion

Summary and relevance

We compared six treatment assignment designs in the detailed grid ranking, plus two related variants and SRS, for a spatial cluster randomized trial on a grid of 100 clusters, under three mechanisms of heterogeneous baseline incidence, two spillover regimes, and a range of values for

spatial dependence, spillover magnitude, and the treatment effect. Some findings held across the incidence configurations and the values of ρ, γ, and τ under queen contiguity. Others depended on the spillover regime, which a trial team may not know in advance.

We judged each design on two criteria: its average MSE, and the spread of MSE across the allocations it can produce. A trial realizes one allocation, so a design with a low average MSE can still give a poor estimate if some of its allocations perform badly. Chapter 2 compared simple random and block stratified sampling on the same two criteria. The spread matters most for SRS under control-only spillover: in the grid pilot, the worst 10% of SRS allocations had mean conditional MSE 0.3486, against 0.2172 for SRS allocations overall.

Under queen contiguity, Incidence-Guided Saturation had the lowest mean squared error (0.091) and Balanced Quartiles the second lowest (0.131), and the same two designs led under rook contiguity (0.072 and 0.084). Under queen contiguity, the order of the six designs was the same in each of the five incidence configurations and at every value of τ, and neither the strength of spatial dependence in the outcome nor the spillover magnitude changed it; the mechanism that generated the incidence surface changed each design's MSE modestly (by under 10%, except High Incidence Focus). Coverage of the 95% intervals was close to 0.94 for every design under both neighbor types, except Checkerboard under rook contiguity, so the designs differed in precision and coverage did not separate them.

The spillover regime was the one factor that changed the queen ranking. Under queen contiguity at $\tau = 1$, Checkerboard and 2×2 Blocking were close to the best designs when spillover reached both arms and were the two worst when it reached only control clusters (MSE 0.062 and 2.111 for Checkerboard, 0.062 and 0.576 for 2×2 Blocking), and Balanced Quartiles also lost ground under the control-only regime (0.042 and 0.219), while Incidence-Guided Saturation was first or second among the six detailed designs under both regimes at every value of τ and remained competitive with SRS under both-arms spillover. Its control-only MSE did rise with τ, and

Isolation Buffer, whose MSE did not, matched it under that regime at $\tau = 2$ and passed it at $\tau = 3$. A design whose performance depends on the regime carries a risk that its average MSE does not show.

Checkerboard, the strict allocation of Chapter 2, failed for different reasons under the two neighbor types. Under queen contiguity its estimates stayed centered on τ with close to nominal coverage but were imprecise, because under the control-only regime only the boundary clusters separate the spillover effect from the treatment effect; under rook contiguity τ is not identified, and its estimates were centered on $\tau - \gamma$. Checkerboard intersperses treated and control clusters as finely as the grid allows, and both failures come from the dependence this pattern creates between Z and the spillover covariate (Methods).

Together, the grid benchmark, the allocation-risk results (Table 9) and the NC application give guidance by scenario. When spillover is expected to reach mainly control clusters, Incidence-Guided Saturation and Saturation Quadrants had the lowest mean MSE and among the lowest worst-decile MSE, so they are the designs we recommend over SRS. Isolation Buffer also limited downside in that regime, but it should be considered only when both-arms spillover is implausible and a smaller treated budget is acceptable. When spillover is expected to reach both arms, the spread of MSE across allocations was small for SRS and the balancing and saturation designs, and SRS, Balanced Quartiles, Balanced Halves and the saturation designs performed about equally, so SRS remains a reasonable choice. High Incidence Focus gave one fixed allocation whose MSE in NC was below the SRS worst-decile mean under both regimes. Its accuracy is model-based rather than randomization-based, however: it was about four times worse than SRS under both arms in the grid pilot, and its MSE rose 1.4–4.4-fold when baseline outcomes were related to incidence. It is a candidate only when historical incidence can be assumed unrelated to the outcome. Checkerboard and 2×2 Blocking did not improve on SRS on either criterion. The NC model assesses education outcomes rather than assuming that historical SUD incidence

predicts education ability or a heterogeneous intervention effect.

Saturation Quadrants, which assigns the same saturations without reference to incidence, performed similarly to Incidence-Guided Saturation (see Treatment assignment designs), so their accuracy advantage over SRS under control-only spillover comes from the shared saturation structure, and incidence guidance lets a trial direct its highest treatment saturations to the highest-incidence regions with little or no loss of estimation accuracy.

These are design-stage results, and they complement analysis-stage modeling rather than replace it. Every design was analyzed with a spatial lag model that included the spillover term, and, in the grid study, for every design whose τ was identified the estimates were close to unbiased: at $\tau = 1$, the absolute bias of each design under each regime was at most 0.048 under queen contiguity and 0.027 under rook contiguity. Given that model, the choice of design determined how precisely τ was estimated.

Relation to Chapter 2

Chapter 2's practical argument considered reasonably good estimation together with protection against poor allocation in its studied small-grid settings. We preserve both criteria while extending the designs, geography and incidence conditions. Our findings concern these new models and settings; they do not directly compare numerical results across chapters. The rectangular Checkerboard and its graph adaptation illustrate why a restricted allocation's low assignment variation must be interpreted together with its level of estimation error. The application also separates incidence-guided assignment from assumptions about education outcomes. No direct cross-chapter result comparison is needed to motivate these extensions.

Limitations

These conclusions have several limitations. First, the primary estimator is an oracle. It includes the exact spillover exposure term, so its results are a best case for each design. We fit a non-oracle

model that omits the spillover term as a sensitivity analysis; it was biased for every design and its intervals under-covered, but Incidence-Guided Saturation still had the lowest MSE under both neighbor types. The non-oracle model is only one departure from the oracle. We did not study an analyst who includes a spillover term of the wrong form, such as one built on the wrong neighbor definition or a distance decay.

Second, in the synthetic grid study, the designs and the fitted model used the incidence surface that generated the outcome (the matched-surface assumption in Methods). In practice, planning incidence is estimated from an earlier period and measured with error. The designs that assign treatment by incidence rank, and the covariate adjustment, would then use a noisy surface, and we did not study how that changes the comparison.

Third, in the Poisson mechanism every cluster had the same population of 100,000. Real clusters, including the community college service areas in the Application, differ in population, so their observed rates differ in precision, and unequal populations may change how the designs that use incidence perform.

Fourth, the full synthetic comparison uses one regular 10×10 grid, with β and σ fixed at 1. The application adds one irregular 58-cluster geography and unequal observed populations, but does not constitute a systematic sweep over graph sizes, structures or participant budgets. Some grid results depend on its boundary geometry, while application design adaptations also change assignment support.

Fifth, the simulation shares random draws by design. Blocks within an incidence configuration share its ten incidence surfaces, so the block-level tests treat dependent blocks as independent, which is why we also compared adjacent designs over the 50 independent configuration-by-surface units. The five values of τ also share their draws, so the stability of the design order across τ is not an independent replication at each value (see Stability across treatment effect sizes).

Sixth, the simulation's base rate of 35 per 100,000 is not the rate of the county data described

in the Application (see Heterogeneous baseline incidence); we kept it deliberately, because the simulation is a general design study rather than a calibration to one data set. In the Poisson mechanism, incidence enters the designs and the fitted model only through the ranks of the observed rates, so the base rate affects the simulation only through the spatial pattern of those ranks and the number of tied rates.

Seventh, application results are conditional on a continuous education-outcome model with known spillover form and constant structural effect, not an observed education intervention or demonstrated reduction in SUD. Working-age population is a geographic denominator rather than a sample of healthcare professionals. Historical rates are held fixed, so uncertainty in their measurement and changes between planning and implementation are not propagated. The matched baseline sensitivity is hypothetical.

Eighth, 95% Wald coverage remains below nominal in both the identified grid settings and the application. Greater Monte Carlo precision does not repair this calibration. The finite-outcome allocation-tail estimates do not remove noise from individual risks or establish population worst cases. Settings and years that share the same simulations cannot be treated as independent confirmation. Budgets also differ; these are evaluations of retained strategies, not an equal-cost optimization.

Future work

The next trial-design steps are to specify the education endpoint, eligible professionals, implementation budget and plausible spillover network, then assess interval calibration for that protocol before randomization. Additional methodological work could study noisy planning incidence, effect heterogeneity, misspecified exposure networks, unequal participant precision and a broader set of graph geometries. A larger allocation sample could sharpen downside estimates if that precision is needed for a final choice. These are extensions of the completed application, not missing analyses required to characterize the prespecified settings reported here.

References

[1] Allan Donner and Neil Klar. *Design and analysis of cluster randomization trials in health research*. Arnold, 2000.

[2] Richard J. Hayes and Lawrence H. Moulton. *Cluster Randomised Trials, Second Edition*. Chapman and Hall/CRC, 2017.

[3] Donald B. Rubin. Statistics and causal inference: Comment: Which ifs have causal answers. *Journal of the American Statistical Association*, 81(396):961–962, 1986.

[4] Michael G Hudgens and M. Elizabeth Halloran. Toward Causal Inference With Interference. *Journal of the American Statistical Association*, 103(482):832–842, 2008.

[5] W. R. Tobler. A Computer Movie Simulating Urban Growth in the Detroit Region. *Economic Geography*, 46:234, 1970.

[6] F. N. Binka, F. Indome, and T. Smith. Impact of spatial distribution of permethrin-impregnated bed nets on child mortality in rural northern Ghana. *The American Journal of Tropical Medicine and Hygiene*, 59(1):80–85, 1998.

[7] Edward Miguel and Michael Kremer. Worms: Identifying Impacts on Education and Health in the Presence of Treatment Externalities. *Econometrica*, 72(1):159–217, 2004.

[8] P. A. P. Moran. The Interpretation of Statistical Maps. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 10(2):243–251, 1948.

[9] James LeSage and Robert Kelley Pace. *Introduction to Spatial Econometrics*. Chapman and Hall/CRC, 2009.

[10] Keith Ord. Estimation Methods for Models of Spatial Interaction. *Journal of the American Statistical Association*, 70(349):120–126, 1975.

[11] Roger Bivand. R Packages for Analyzing Spatial Data: A Comparative Case Study with Areal Data. *Geographical Analysis*, 54(3):488–518, 2022.

[12] Christopher Jarvis, Gian Luca Di Tanna, Daniel Lewis, Neal Alexander, and W. John Edmunds. Spatial analysis of cluster randomised trials: a systematic review of analysis methods. *Emerging Themes in Epidemiology*, 14(1):12, 2017.

[13] Gillian M. Raab and Izzy Butcher. Balance in cluster randomized trials. *Statistics in Medicine*, 20(3):351–365, 2001.

[14] Lawrence H. Moulton. Covariate-based constrained randomization of group-randomized trials. *Clinical Trials*, 1(3):297–305, 2004.

[15] Amy M Crisp, M Elizabeth Halloran, Ira M Longini, Gonzalo Vazquez-Prokopec, and Natalie E Dean. Covariate-constrained randomization with cluster selection and substitution. *Clinical Trials*, 20(3):284–292, 2023.

[16] Sarah Baird, J. Aislinn Bohren, Craig McIntosh, and Berk Özler. Optimal design of experiments in the presence of interference. *The Review of Economics and Statistics*, 100(5):844–860, 2018.

[17] Robert S. McCann, Henk van den Berg, Willem Takken, Andrew G. Chetwynd, Emanuele Giorgi, Dianne J. Terlouw, and Peter J. Diggle. Reducing contamination risk in cluster-randomized infectious disease-intervention trials. *International Journal of Epidemiology*, 47(6):2015–2024, 2018.

[18] Michael P. Leung. Rate-optimal cluster-randomized designs for spatial interference. *The Annals of Statistics*, 50(5):3064–3087, 2022.

[19] Chencheng Cai, Xu Zhang, and Edoardo M. Airoldi. Independent-set design of experiments for estimating treatment and spillover effects under network interference. In *Proceedings*

of the 12th International Conference on Learning Representations (ICLR 2024), Vienna, Austria, 2024.

[20] Michael P. Leung. Cluster-randomized trials with cross-cluster interference. *Journal of the American Statistical Association*, pages 1361–1371, 2025.

[21] Samuel I. Watson and Thomas A. Smith. Design and Analysis of Randomized Trials to Estimate Spatio-Temporally Heterogeneous Treatment Effects. *Journal of the American Statistical Association*, 121(553):502–512, 2026.

[22] Mojtaba Mirzaei, Golsa Joodi, Brittany Bogle, Sarah Chen, and Ross J. Simpson. Years of Life and Productivity Loss Because of Adult Sudden Unexpected Death in the United States. *Medical Care*, 57(7):498–502, 2019.

[23] Roger Bivand and Gianfranco Piras. Comparing Implementations of Estimation Methods for Spatial Econometrics. *Journal of Statistical Software*, 63(18), 2015.

[24] Milton Friedman. The use of ranks to avoid the assumption of normality implicit in the analysis of variance. *Journal of the American Statistical Association*, 32(200):675–701, 1937.

[25] Peter Bjorn Nemenyi. *Distribution-Free Multiple Comparisons*. PhD thesis, Princeton University, Princeton, NJ, 1963.

[26] Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006.

[27] Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.

[28] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

[29] Ashkan Habib. Temporal trends and social determinants of sudden death mortality in

North Carolina. Master's paper, Department of Biostatistics, University of North Carolina at Chapel Hill, Chapel Hill, NC, April 2026. Unpublished.

Appendix

A1. The eight-design comparison

Table 10 defines the two related variants excluded from the detailed six-design grid ranking but included in the main-text nine-design benchmark and application, in the format of Table 1, and Table 11 reports the comparison of all eight designs at the primary scenario, $\tau = 1$.

Table 10: The two related variants outside the detailed six-design grid ranking, in the format of Table 1. Saturation Quadrants shares its mechanism with Incidence-Guided Saturation, and Balanced Halves with Balanced Quartiles (see Treatment assignment designs). A cluster's incidence rank is its rank by the baseline incidence surface X_k, with ties broken at random in every draw. Clusters treated is the number of clusters treated per assignment.

Design	Assignment rule	Clusters treated	Deterministic
Saturation Quadrants	The grid is divided into its four 5×5 coordinate quadrants, which are given saturations of 20%, 40%, 60%, and 80% in an order drawn at random for each assignment, without reference to incidence; within each quadrant, that share of its 25 clusters is treated at random	50	No
Balanced Halves	Clusters are divided by incidence rank into two strata of 50; 25 clusters are treated at random in each stratum	50	No

Each excluded design was compared with its retained counterpart by the Nemenyi and Holm-adjusted Wilcoxon signed-rank tests described under Simulation design, resampling, and metrics, applied here to all eight designs. Saturation Quadrants's mean squared error was 0.0900, against 0.0907 for Incidence-Guided Saturation, under queen contiguity (Nemenyi $p = 0.98$; Holm-adjusted Wilcoxon signed-rank $p = 0.41$), and 0.0708 against 0.0717 under rook contiguity (Nemenyi $p = 1.00$; Wilcoxon $p = 0.82$). Saturation Quadrants had the marginally lower mean under both neighbor types and the better average rank in the eight-design comparison (2.100 against 2.350 under queen contiguity and 2.281 against 2.306 under rook contiguity), so retaining Incidence-Guided Saturation as the representative of the saturation mechanism does not rest on performance. Balanced Halves had a mean squared error of 0.1299, against 0.1308 for Balanced Quartiles, under queen contiguity (Nemenyi $p = 1.00$; Wilcoxon $p = 0.88$), and 0.0842 against

0.0837 under rook contiguity (Nemenyi $p = 1.00$; Wilcoxon $p = 0.82$). The two designs differ only in how finely they stratify on incidence, and we retain the finer, four-stratum version. Balanced Halves had the marginally lower mean and the better average rank under queen contiguity (3.312 against 3.356), and Balanced Quartiles had both under rook contiguity (2.919 against 3.038). Under both neighbor types, each excluded design is adjacent to its retained counterpart in mean MSE (Table 11). The MSEs and coverages of the six retained designs are those of Table 2, given here to four and three decimal places.

Table 11: Performance of all eight designs at the primary scenario, $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages the 160 scenarios of one design and neighbor type (5 incidence configurations $\times$ 4 values of ρ $\times$ 4 values of γ $\times$ 2 spillover regimes), each summarizing 250 fits of the oracle model. Rows in both panels are in order of mean squared error (MSE) under queen contiguity. Retained marks the six designs compared in the body of this chapter. Average rank is the mean of each design's within-block MSE rank among the eight designs over the 160 blocks (1 = lowest MSE), so it differs from the six-design average rank in Table 2. Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.

Design	Retained	MSE	Coverage	Average rank
Queen contiguity (primary)				
Saturation Quadrants	No	0.0900	0.936	2.100
Incidence-Guided Saturation	Yes	0.0907	0.937	2.350
Balanced Halves	No	0.1299	0.936	3.312
Balanced Quartiles	Yes	0.1308	0.936	3.356
Isolation Buffer	Yes	0.1602	0.939	5.138
High Incidence Focus	Yes	0.2455	0.938	6.763
2×2 Blocking	Yes	0.3192	0.939	6.244
Checkerboard	Yes	1.0869	0.941	6.737
Rook contiguity (sensitivity)				
Saturation Quadrants	No	0.0708	0.940	2.281
Incidence-Guided Saturation	Yes	0.0717	0.942	2.306
Balanced Halves	No	0.0842	0.940	3.038
Balanced Quartiles	Yes	0.0837	0.940	2.919
Isolation Buffer	Yes	0.1329	0.939	5.225
High Incidence Focus	Yes	0.2062	0.939	6.969
2×2 Blocking	Yes	0.1281	0.938	5.275
Checkerboard (flagged)	Yes	0.4799	0.150	7.987

A2. Simulation mechanics

This section gives the mechanical details behind Simulation design, resampling, and metrics (Methods): the parameter grid, the random-number streams, the Monte Carlo structure, and the exact formulas for the metrics and their Monte Carlo standard errors.

Parameter grid. Table 12 lists the parameters that were crossed to form the 12,800 scenarios and the settings that were held fixed.

Table 12: Parameter grid of the original eight-design simulation. The crossed parameters form $5 \times 5 \times 2 \times 4 \times 4 \times 2 \times 8 = 12{,}800$ scenarios, 9,600 of them for the six designs compared in the body of this chapter; the settings in the lower panel are the same in every scenario.

Parameter	Values	Levels
Crossed		
Treatment effect τ	0.8, 1.0 (primary), 1.5, 2.0, 3.0	5
Incidence configuration	iid; spatially autocorrelated, $\rho_X \in \{0.20, 0.50\}$; Poisson, $\rho_X \in \{0.20, 0.50\}$	5
Neighbor type	Queen (primary), rook (sensitivity)	2
Outcome spatial dependence ρ	0, 0.01, 0.20, 0.50	4
Spillover magnitude γ	0.5, 0.6, 0.7, 0.8	4
Spillover regime	Both arms, control only (Equation 2)	2
Design	The six designs of Table 1 and the two of Table 10	8
Fixed		
Grid	Regular 10×10 grid, $N = 100$ clusters	
Outcome model (Equation 1)	$\beta = 1$, $\sigma = 1$	
Poisson mechanism	Base rate 35 per 100,000; $P = 100{,}000$ per cluster	
Incidence surfaces per configuration	$K = 10$	
Assignments per surface	$J = 25$ (Checkerboard: one assignment, used in every draw)	
Fits per scenario and estimator	$KJ = 250$, each with its own noise vector	
Estimators	Oracle (primary) and non-oracle (sensitivity), both fit to every simulated outcome	

Random-number streams. Every random draw in the simulation has its own seed, computed by hashing a key that names the draw. The key lists the draw's type (incidence surface, assignments, or noise) and the parameters that identify it, with numeric values written to a fixed number of decimal places so that equal values always give the same key. The random-number generator and its methods for normal and discrete draws are also fixed explicitly, so the streams do not depend on software defaults. Table 13 lists the three kinds of draw, the parameters in their keys, and the scenarios that therefore share them. The consequences of this scheme, independence from the order of execution and common random numbers, are described in Methods.

Monte Carlo structure and checkpoints. Within a scenario, fit (k, j), with $k = 1, \ldots, 10$ and $j = 1, \ldots, 25$, uses surface X_k, assignment Z_{kj}, and noise vector ϵ_{kj}, the (k, j) column of the block's 100×250 noise matrix (see Simulation design, resampling, and metrics). The unit of

Table 13: Seed keys. Each key names the kind of draw and the parameters listed here; the incidence mechanism is iid, spatial, or Poisson, and the iid configuration, which has no ρ_X, uses 0 in its place. k indexes the incidence surface and d the design. A draw is shared by every scenario whose key parameters match its own.

Draw	Key parameters	Shared by
Incidence surface X_k	Incidence mechanism, ρ_X, k	Every block, design, and value of τ of the configuration
The $J = 25$ assignments of design d on surface k	Incidence mechanism, ρ_X, neighbor type, ρ, γ, spillover regime, k, d	All five values of τ and both estimators
The 250 noise vectors of a block	Incidence mechanism, ρ_X, neighbor type, ρ, γ, spillover regime	All eight designs, all five values of τ, and both estimators

computation is a combination of incidence configuration, neighbor type, and ρ: 40 units, each covering 4 values of γ, 2 spillover regimes, 8 designs, 5 values of τ, 250 fits, and 2 estimators. The units ran in parallel, and the run stops if any unit fails. Each unit's results are saved together with a record of the parameter settings, the simulation code, and the software versions that produced them, and saved results are reused only if all of these match the current run, so results from different code or settings are never combined.

Scenario metrics. For one scenario and estimator, let $f = 1, \dots, n$ index the valid fits, those with a finite estimate and standard error (here every fit was valid, so $n = 250$). Let $L_f, U_f = \hat{\tau}_f \mp z_{0.975}\,\widehat{\mathrm{SE}}_f$, where $z_{0.975} = \Phi^{-1}(0.975) \approx 1.96$, and let $\bar{\hat{\tau}} = n^{-1}\sum_f \hat{\tau}_f$. Then

$$\mathrm{Bias} = \frac{1}{n}\sum_{f=1}^{n}(\hat{\tau}_f - \tau), \qquad \mathrm{SD} = \left\{\frac{1}{n-1}\sum_{f=1}^{n}\left(\hat{\tau}_f - \bar{\hat{\tau}}\right)^2\right\}^{1/2}, \qquad \mathrm{MSE} = \frac{1}{n}\sum_{f=1}^{n}(\hat{\tau}_f - \tau)^2,$$

$$\mathrm{Coverage} = \frac{1}{n}\sum_{f=1}^{n}\mathbf{1}\{L_f \le \tau \le U_f\}, \qquad \mathrm{Power} = \frac{1}{n}\sum_{f=1}^{n}\mathbf{1}\{L_f > 0\},$$

so that $\mathrm{MSE} = \mathrm{Bias}^2 + (n-1)\,\mathrm{SD}^2/n$.

Surface-level standard errors. For $m \in \{\mathrm{Bias}, \mathrm{MSE}, \mathrm{Coverage}, \mathrm{Power}\}$, let m_k be the same formula applied to the valid fits from surface k (here all 25). With $K = 10$ and $s(a_1, \dots, a_K) =$

$\{(K-1)^{-1}\sum_k (a_k-\bar{a})^2\}^{1/2}$ the sample standard deviation,

$$\mathrm{SE}(m) = \frac{s(m_1,\dots,m_K)}{\sqrt{K}}, \qquad m \pm t_{0.975,\,K-1}\,\mathrm{SE}(m).$$

Because every fit was valid, each surface contributes 25 fits and $m = K^{-1}\sum_k m_k$. For two designs d and d' in the same block at the same τ, the difference is paired at the surface level: with $\Delta_k = m_{d,k} - m_{d',k}$,

$$\mathrm{SE}(m_d - m_{d'}) = \frac{s(\Delta_1,\dots,\Delta_K)}{\sqrt{K}},$$

with the same $t_{0.975,\,K-1}$ interval.

Standard errors pooled over configurations. For one design, neighbor type, and value of τ, let $\mathcal{B}_c$ be the 32 blocks of configuration c and $m_{b,k}$ the surface-k metric in block b. The configuration-by-surface means and the pooled metric are

$$\bar{m}_{c,k} = \frac{1}{32}\sum_{b\in\mathcal{B}_c} m_{b,k}, \qquad \bar{m} = \frac{1}{50}\sum_{c=1}^{5}\sum_{k=1}^{10}\bar{m}_{c,k},$$

where $\bar{m}$ equals the mean of the metric over the 160 scenarios, and

$$\mathrm{SE}(\bar{m}) = \frac{s(\{\bar{m}_{c,k}\})}{\sqrt{50}},$$

with s taken over the 50 values $\bar{m}_{c,k}$. These are the standard errors in Table 2. For two designs d and d' adjacent in the mean-MSE order (Design performance under queen contiguity), let $D_{c,k} = \bar{m}_{d,c,k} - \bar{m}_{d',c,k}$ for $m = \mathrm{MSE}$. The interval for the mean difference is $\bar{D} \pm t_{0.975,\,49}\, s(\{D_{c,k}\})/\sqrt{50}$, and the Wilcoxon signed-rank test, with the normal approximation, is applied to the 50 values $D_{c,k}$.

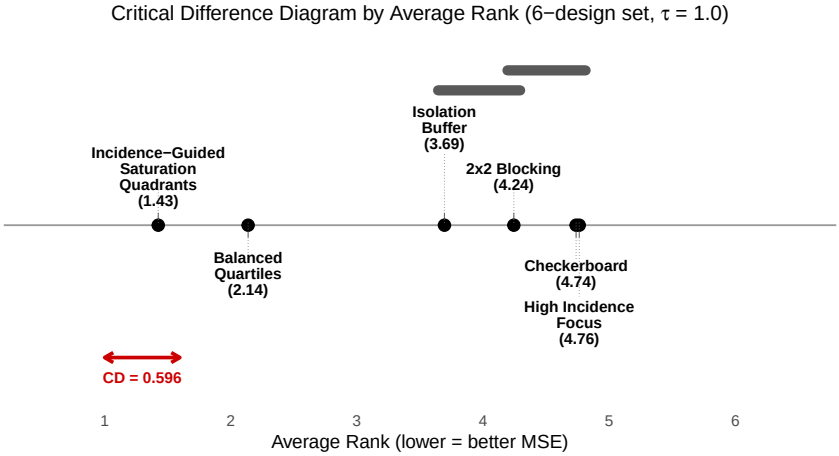
A3. Supplementary statistical comparisons

This section gives the pairwise test results and two figures behind results stated in the body of this chapter.

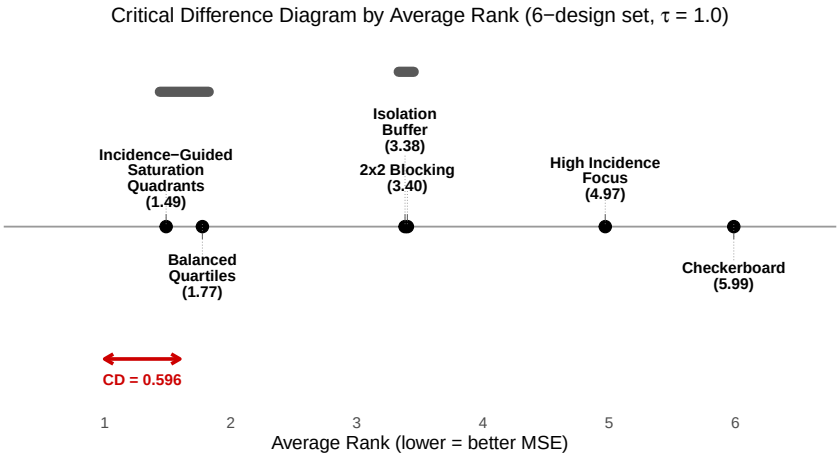
Pairwise tests under queen contiguity. The four pairs that Nemenyi comparisons did not separate were 2×2 Blocking against Isolation Buffer ($p = 0.090$), High Incidence Focus ($p = 0.130$), and Checkerboard ($p = 0.170$), and Checkerboard against High Incidence Focus ($p = 1.00$). Over the 50 independent configuration-by-surface units, the mean paired differences in MSE between designs adjacent in the mean-MSE order, with t_{49} 95% intervals, were -0.0401 (-0.0431, -0.0372) for Incidence-Guided Saturation against Balanced Quartiles, -0.0293 (-0.0332, -0.0255) for Balanced Quartiles against Isolation Buffer, -0.0853 (-0.0954, -0.0752) for Isolation Buffer against High Incidence Focus, -0.0738 (-0.0858, -0.0617) for High Incidence Focus against 2×2 Blocking, and -0.7677 (-0.7893, -0.7460) for 2×2 Blocking against Checkerboard, with Wilcoxon $p \leq 4.9 \times 10^{-9}$ in every case. The better design of each pair had the lower MSE in all 50 units, except High Incidence Focus against 2×2 Blocking, where it did so in 48 (96%). This test follows the mean-MSE order. It separated High Incidence Focus from 2×2 Blocking, which the Nemenyi comparison of within-block ranks did not, because each unit's MSE averages over both spillover regimes, and there the control-only losses of 2×2 Blocking outweigh its both-arms wins.

Pairwise tests under rook contiguity. Nemenyi comparisons separated 13 of 15 pairs; the two they did not separate were Incidence-Guided Saturation against Balanced Quartiles ($p = 0.74$) and 2×2 Blocking against Isolation Buffer ($p = 1.00$). Holm-adjusted Wilcoxon tests separated every pair except 2×2 Blocking against Isolation Buffer. In the surface-level comparison, Incidence-Guided Saturation had a lower MSE than Balanced Quartiles in 48 of the 50 units (mean difference -0.0121; -0.0139, -0.0102; Wilcoxon $p = 9.9 \times 10^{-10}$). 2×2 Blocking and Isolation Buffer were close: 2×2 Blocking had the lower MSE in 36 of the 50 units

(72%; mean difference −0.0048; −0.0078, −0.0017; Wilcoxon $p = 0.0027$), but Isolation Buffer had the slightly better average rank (3.38 against 3.40), and the two designs traded places at some values of τ. Every other adjacent pair differed in all 50 units. Within single incidence configurations, rook coverage was between 0.935 and 0.946 for every design except Checkerboard, whose coverage was between 0.14 and 0.15.



(a) Queen contiguity (primary)

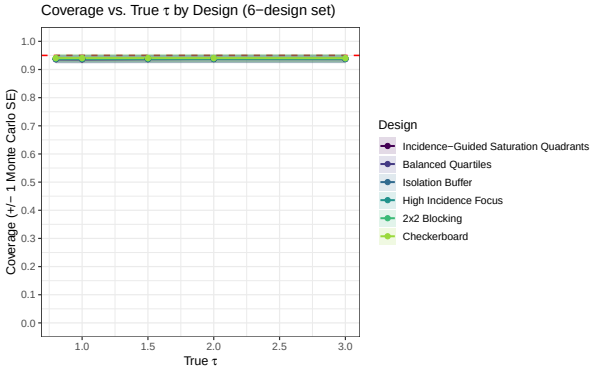


(b) Rook contiguity (sensitivity)

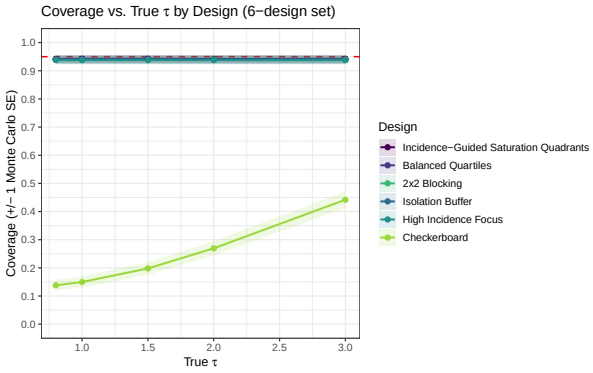
Figure 9: Nemenyi critical-difference diagrams for the six designs at $\tau = 1$, pooled over the five incidence configurations. Designs are placed at their average within-block MSE rank, given in parentheses (1 = lowest MSE); gray bars join designs that the Nemenyi comparisons did not separate.

Figure 9 gives the Nemenyi critical-difference diagrams for the comparison of the six designs at $\tau = 1$, whose results are reported under Design performance under queen contiguity and Rook contiguity in the Results. Each design is placed at its average within-block MSE rank among the

six designs over the 160 blocks of its neighbor type. The red arrow shows the critical difference, 0.596 at $\alpha = 0.05$, and a gray bar joins designs whose pairwise Nemenyi comparisons were all not significant: under queen contiguity, Isolation Buffer with 2×2 Blocking, and 2×2 Blocking with Checkerboard and High Incidence Focus; under rook contiguity, Incidence-Guided Saturation with Balanced Quartiles, and Isolation Buffer with 2×2 Blocking. Where two designs have nearly equal average ranks (Checkerboard and High Incidence Focus under queen contiguity, Isolation Buffer and 2×2 Blocking under rook contiguity), the dotted leader line to the outer label crosses the neighboring design's label; the positions and labels are correct. Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.



(a) Queen contiguity (primary)



(b) Rook contiguity (sensitivity)

Figure 10: Coverage of the 95% confidence intervals by design across $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$, pooled over the five incidence configurations. Each point averages the 160 scenarios of one design at one value of τ. The dashed red line marks the nominal 0.95 level.

Figure 10 gives the coverage of each design across the five values of τ, summarized under Stability across treatment effect sizes. The shaded bands show ± 1 times the average of the

scenario-level Monte Carlo standard errors of coverage; they describe the precision of a single scenario's coverage, not of the plotted average. The legend lists the designs in order of pooled mean MSE at $\tau = 1$ for the neighbor type of the panel (Table 2). The vertical axis runs from 0 to 1, so lines close to 0.94 overlap: all six under queen contiguity, between 0.936 and 0.941, and all but Checkerboard under rook contiguity, between 0.937 and 0.942. The lowest line under rook contiguity, which the legend does not mark, is Checkerboard, for which τ is not identified; its estimates are kept and flagged. The five values of τ share their assignments and noise.

A4. Additional sensitivity tables

This section gives three tables summarized in the Results: mean squared error by level of outcome spatial dependence ρ (Table 14), Checkerboard's coverage by level of ρ under each neighbor type (Table 15), and mean squared error by level of spillover magnitude γ (Table 16). The first and third are discussed under Sensitivity to spatial dependence, spillover magnitude, and spillover regime, and the second under Rook contiguity.

Table 14: Mean squared error by design across the four levels of outcome spatial dependence ρ, under queen contiguity at $\tau = 1$, pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations $\times$ 4 values of γ $\times$ 2 spillover regimes). Rows are ordered by pooled mean MSE under queen contiguity (Table 2).

Design	$\rho = 0$	$\rho = 0.01$	$\rho = 0.20$	$\rho = 0.50$
Incidence-Guided Saturation	0.090	0.087	0.093	0.093
Balanced Quartiles	0.129	0.129	0.132	0.133
Isolation Buffer	0.161	0.157	0.158	0.165
High Incidence Focus	0.243	0.242	0.249	0.248
2×2 Blocking	0.318	0.320	0.314	0.324
Checkerboard	1.095	1.077	1.084	1.092

Table 15: Checkerboard's 95% confidence interval coverage across the four levels of outcome spatial dependence ρ at $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations $\times$ 4 values of γ $\times$ 2 spillover regimes). Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged. The nominal level is 0.95.

Neighbor type	$\rho = 0$	$\rho = 0.01$	$\rho = 0.20$	$\rho = 0.50$
Queen	0.941	0.943	0.940	0.939
Rook	0.161	0.156	0.148	0.134

Table 16: Mean squared error by design across the four levels of spillover magnitude γ at $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations $\times$ 4 values of ρ $\times$ 2 spillover regimes). Rows in both panels are ordered by pooled mean MSE under queen contiguity (Table 2). Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.

Design	$\gamma = 0.5$	$\gamma = 0.6$	$\gamma = 0.7$	$\gamma = 0.8$
Queen contiguity				
Incidence-Guided Saturation	0.092	0.089	0.092	0.089
Balanced Quartiles	0.132	0.128	0.132	0.131
Isolation Buffer	0.159	0.163	0.160	0.159
High Incidence Focus	0.244	0.249	0.248	0.241
2×2 Blocking	0.312	0.314	0.328	0.323
Checkerboard	1.108	1.052	1.086	1.101
Rook contiguity				
Incidence-Guided Saturation	0.069	0.072	0.075	0.072
Balanced Quartiles	0.083	0.085	0.083	0.083
Isolation Buffer	0.131	0.133	0.134	0.133
High Incidence Focus	0.207	0.209	0.205	0.203
2×2 Blocking	0.129	0.129	0.127	0.128
Checkerboard (flagged)	0.295	0.403	0.538	0.684

A5. Coverage, bias–variance and rook contiguity figures

This section gives two queen-contiguity figures summarized in the Results, the coverage of each design (Figure 11) and the decomposition of MSE into squared bias and variance for one incidence configuration (Figure 12), followed by the rook-contiguity versions of these two figures and of Figure 2, as described under Rook contiguity in the Results. Each rook figure has the layout of its queen counterpart, but designs are ordered for rook contiguity, so the order differs from that of the queen figure. Under rook contiguity, τ is not identified for Checkerboard ($WZ = 1 - Z$); its estimates are kept and flagged.

Figure 13 corresponds to Figure 2.

Figure 14 corresponds to Figure 11. Every design except Checkerboard has coverage close to the nominal level; Checkerboard's coverage, 0.150 pooled over the five incidence configurations (Table 2), is far below it.

Figure 15 corresponds to Figure 12. Nearly all of Checkerboard's MSE is squared bias, because

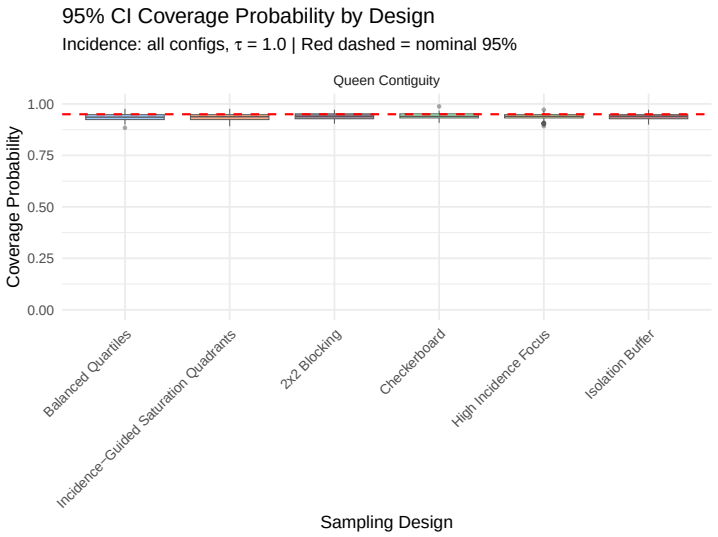


Figure 11: Coverage of the 95% confidence intervals by design at $\tau = 1$, under queen contiguity. Each box summarizes the coverage of the 160 scenarios of one design (5 incidence configurations $\times$ 4 values of ρ $\times$ 4 values of γ $\times$ 2 spillover regimes), each estimated from 250 fits. Designs are ordered by median coverage, so the order differs from the MSE order used elsewhere. The dashed red line marks the nominal 0.95 level. Figure 14 shows rook contiguity.

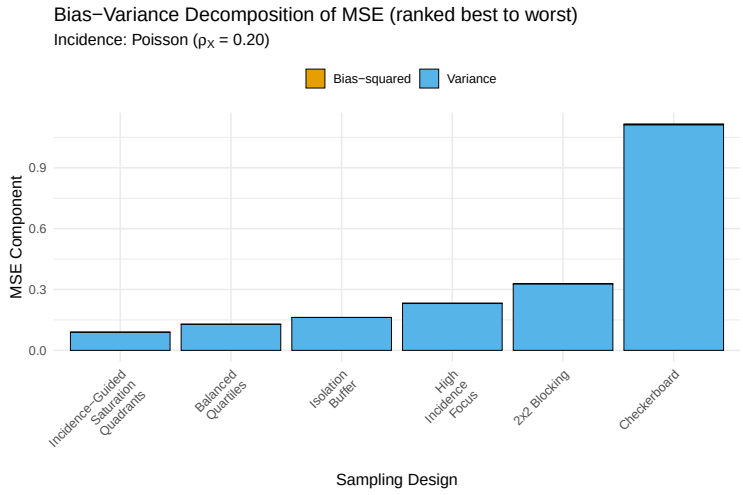


Figure 12: Decomposition of mean squared error into squared bias and variance by design at $\tau = 1$ for one incidence configuration, Poisson with $\rho_X = 0.20$, under queen contiguity. For each design, the two stacked components are the squared bias and the variance (SD^2) of $\hat{\tau}$, each averaged over the 32 scenarios of that configuration (4 values of ρ $\times$ 4 values of γ $\times$ 2 spillover regimes). Designs are ordered by pooled mean MSE (Table 2). Figure 15 shows rook contiguity.

its coefficient on Z estimates $\tau - \gamma$ (Methods).

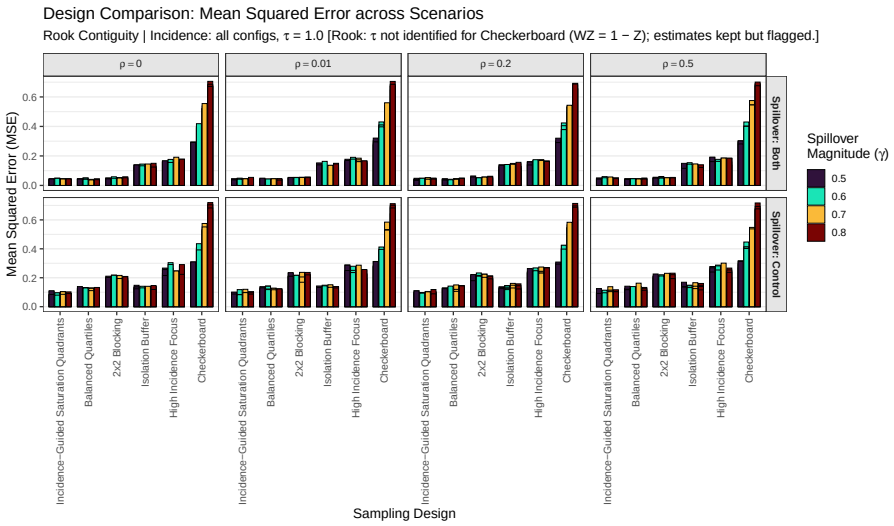


Figure 13: Mean squared error of the oracle estimate of τ by design at $\tau = 1$, under rook contiguity; layout as in Figure 2. Checkerboard is flagged (τ not identified).

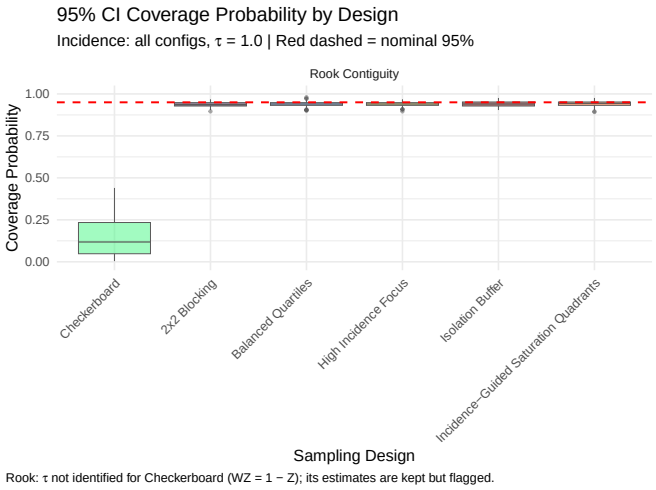


Figure 14: Coverage of the 95% confidence intervals by design at $\tau = 1$, under rook contiguity; layout as in Figure 11. Checkerboard is flagged (τ not identified).

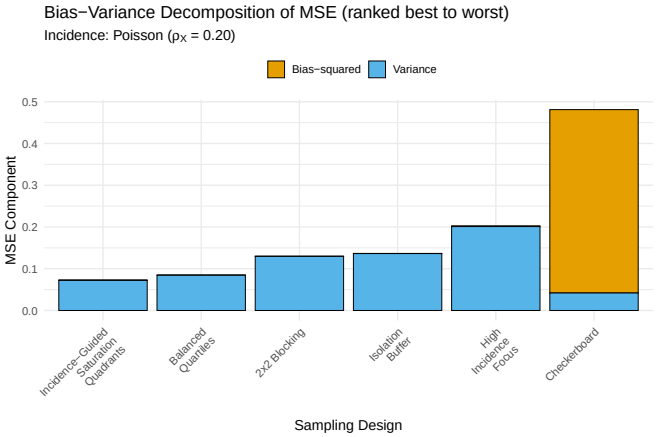


Figure 15: Squared bias and variance of $\hat{\tau}$ by design at $\tau = 1$ for the Poisson configuration with $\rho_X = 0.20$, under rook contiguity; layout as in Figure 12. Checkerboard is flagged (τ not identified).

A6. Oracle and non-oracle estimates

Table 17 gives the mean squared error, bias, and coverage of the oracle and non-oracle models for each of the six designs, as summarized under Non-oracle sensitivity analysis in the Results.

Table 17: Oracle and non-oracle estimates of τ by design at $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages the 160 scenarios of one design and neighbor type. The oracle model has the covariates Z, Spill, and X_k; the non-oracle model omits Spill. Rows in both panels are in order of oracle MSE under queen contiguity. Under rook contiguity, τ is not identified for Checkerboard ($WZ = 1 - Z$); its estimates are kept and flagged, and the two models coincide.

	Oracle			Non-oracle		
Design	MSE	Bias	Coverage	MSE	Bias	Coverage
Queen contiguity (primary)						
Incidence-Guided Saturation	0.091	0.032	0.937	0.087	−0.123	0.810
Balanced Quartiles	0.131	0.017	0.936	0.098	−0.176	0.770
Isolation Buffer	0.160	0.003	0.939	0.091	−0.175	0.879
High Incidence Focus	0.245	0.031	0.938	0.219	−0.180	0.896
2×2 Blocking	0.319	0.010	0.939	0.121	−0.244	0.709
Checkerboard	1.087	0.001	0.941	0.108	−0.206	0.741
Rook contiguity (sensitivity)						
Incidence-Guided Saturation	0.072	0.018	0.942	0.086	−0.121	0.810
Balanced Quartiles	0.084	0.008	0.940	0.097	−0.176	0.772
Isolation Buffer	0.133	0.003	0.939	0.211	−0.400	0.513
High Incidence Focus	0.206	0.018	0.939	0.223	−0.192	0.892
2×2 Blocking	0.128	0.004	0.938	0.128	−0.257	0.691
Checkerboard (flagged)	0.480	−0.652	0.150	0.480	−0.652	0.150

A7. Technical notes

This section explains two results reported in the body of this chapter, the growth with τ of the bias of Incidence-Guided Saturation and the rise with τ of the coverage of Checkerboard under rook contiguity, and closes with block-level detail for the results across τ. The stored results keep neither $\hat{\rho}$ nor the standard error of each fit, so we re-ran every block for the six designs, both neighbor types, and all five values of τ, 2.4 million fits of the oracle model, recording $\hat{\rho}$ and the standard error, and the re-run reproduced the stored results in all 9,600 scenarios. Numbers below that involve $\hat{\rho}$, the standard error, or the terms of Equation 4 come from this re-run; the others come from the stored results.

Identification diagnostics

Two combinations of design and neighbor type weaken or remove the information that separates τ from the spillover term. Under rook contiguity, Checkerboard gives $WZ = 1 - Z$ exactly, because every neighbor of a treated cluster is a control and every neighbor of a control is treated. Both spillover covariates then equal $\gamma(1 - Z)$, a linear combination of the intercept and Z, so τ is not identified: the fitting routine drops the aliased spillover column, and the coefficient on Z estimates $\tau - \gamma$. Chapter 2 noted the same consequence of exact collinearity between treatment and spillover exposure, $\hat{\beta} \approx \beta - \psi$ in its notation. The non-oracle model has no spillover column to drop, but its Checkerboard estimates under rook contiguity have the same defect. Under queen contiguity, each of the 64 interior clusters has four treated and four control neighbors, so $WZ = 1/2$ at every interior cluster. Under the control-only regime the spillover covariate then equals $(\gamma/2)(1 - Z)$ at those clusters, a linear combination of the intercept and Z, and only the 36 clusters on the grid boundary carry information that separates it from the intercept and Z; τ is identified, but imprecisely. Under the both-arms regime it is the constant $\gamma/2$ at interior clusters, confounded with the intercept but not with Z. Isolation Buffer under rook contiguity can also, rarely, draw exactly the checkerboard pattern, a maximal independent set that equals one color class of the grid. This happened in 4 of its 40,000 rook draws and affected one fit in each of 20 scenarios, 4 at each value of τ. We flag every fit in which the fitting routine reports an aliased column and every scenario in which some draw has $\text{rank}([1, Z, WZ]) < 3$. Aliased estimates are kept in all summaries. In the exhibits, we flag Checkerboard under rook contiguity, where every fit is aliased; the Isolation Buffer scenarios, each with one aliased fit among its 250, are noted here and carry no flag in the exhibits.

Estimation error and its dependence on τ

For one fit (k, j), write Y, Z, and ϵ for Y_{kj}, Z_{kj}, and ϵ_{kj} (Simulation design, resampling, and metrics). Let $V = [1, Z, S(Z), X_k]$ be the 100×4 covariate matrix of the oracle model, so

that Equation 1 reads $(I-\rho W)Y = V\theta + \epsilon$ with $\theta = (0, \tau, 1, \beta)^\top$. Let $A = (I-\rho W)^{-1}$ and $V^+ = (V^\top V)^{-1}V^\top$, and for a vector v write $[V^+ v]_Z$ for the coefficient on Z when v is regressed on the columns of V by least squares. Given $\hat{\rho}$, the estimated coefficients are $V^+(I-\hat{\rho}W)Y$ (Estimation). Because $(I-\hat{\rho}W)Y = V\theta + \epsilon + (\rho-\hat{\rho})WY$ and $WY = WA(V\theta+\epsilon)$,

$$\hat{\tau} - \tau = [V^+\epsilon]_Z + (\rho-\hat{\rho})\,h + \tau(\rho-\hat{\rho})\,g, \tag{4}$$

where

$$g = [V^+WAZ]_Z, \qquad h = [V^+WA\{S(Z) + \beta X_k + \epsilon\}]_Z.$$

The first term has mean zero given Z and X_k. The last term carries a factor of τ. It is present exactly when $g \neq 0$, where g is the coefficient on Z when the lagged treatment $WAZ = WZ + \rho W^2 Z + \rho^2 W^3 Z + \cdots$ is regressed on the model's own covariates, and its sign and size are set by the error in $\hat{\rho}$. The concentrated likelihood depends on Y only through the residual sum of squares of $(I-\tilde{\rho}W)Y$ on V, and the part of that residual that involves τ is $\tau(\rho-\tilde{\rho})$ times the residual of WAZ on V. If WAZ lies in the column space of V, that part vanishes for every $\tilde{\rho}$, so $\hat{\rho}$ is the same at every value of τ; if in addition $g = 0$, then $\hat{\tau} - \tau$ is the same at every value of τ. Equation 4 held on every re-run fit, with a largest residual of 4.2×10^{-13}; for Checkerboard under rook contiguity it held in the reduced form given below.

Two consequences of Equation 4 explain the results for τ reported in the body.

First, for designs whose treated clusters have treated neighbors, the last term gives a small positive bias that grows with τ. Under queen contiguity, pooled over the 160 blocks, the bias of Incidence-Guided Saturation rose from 0.029 at $\tau = 0.8$ to 0.047 at $\tau = 3$ (Stability across treatment effect sizes), while that of Isolation Buffer stayed between 0.002 and 0.004. Almost all of the rise is the last term of Equation 4. In our fits, $\hat{\rho}$ was biased downward by about 0.06 on average (for Incidence-Guided Saturation, by 0.071 at $\tau = 0.8$ and 0.054 at $\tau = 3$). We observed

this downward bias in these simulations; we did not derive it or check it against a published finite-sample result. With $\rho - \hat{\rho}$ positive on average, the last term is positive and grows with τ for any design with $g > 0$. For Incidence-Guided Saturation, g averaged 0.371 over the queen fits, and the last term contributed 0.021 of the bias at $\tau = 0.8$ and 0.043 at $\tau = 3$; the first term contributed 0.001 at both values, and the second 0.007 and 0.002. For Isolation Buffer, g averaged 0.020, and the last term contributed 0.001 and 0.004. Most of the difference in g comes from the control-only regime. Under the both-arms regime with $\rho = 0$, the lagged treatment is $WZ = S(Z)/\gamma$, one of the covariates, so $g = 0$ for every design. Under the control-only regime, $WZ = S(Z)/\gamma + Z \circ WZ$, and at $\rho = 0$, g is close to the average share of treated neighbors among treated clusters. That share is 0.57 for Incidence-Guided Saturation, whose treated clusters are concentrated in the high-saturation quadrants, and exactly 0 for Isolation Buffer, whose treated clusters have no treated neighbors. For Isolation Buffer, $Z \circ WZ = 0$, so the two regimes define the same spillover covariate and the same g, which is 0 at $\rho = 0$ and about 0.06 at $\rho = 0.5$; at $\rho = 0$, its $\hat{\rho}$ and its estimation error were the same at every value of τ, up to the optimizer's tolerance.

The same term explains why the control-only MSE of Incidence-Guided Saturation rose from 0.131 at $\tau = 0.8$ to 0.182 at $\tau = 3$ while that of Isolation Buffer did not. Under that regime, averaged over the 80 queen blocks, the variance of the last term of Equation 4 rose from 0.008 to 0.061 for Incidence-Guided Saturation, while the variance of the rest of the error changed little (0.119 and 0.115). The variance of the last term grew more slowly than τ^2 because $\hat{\rho}$ became less variable as τ grew. For Isolation Buffer, the variance of the last term stayed below 0.0002. The rise in bias slowed at larger values of τ for a related reason: the part of the likelihood that involves τ sharpens it around the true ρ as τ grows, so the downward bias of $\hat{\rho}$ shrank.

Second, the decomposition explains why the coverage of Checkerboard under rook contiguity rose with τ. Under rook contiguity, Checkerboard's coverage rose from 0.138 at $\tau = 0.8$ to

0.442 at $\tau = 3$ (Stability across treatment effect sizes), although its coefficient on Z estimates $\tau - \gamma$ at every value of τ. For this design $WZ = 1 - Z$, and because W is row-standardized, $W^2Z = W(1 - Z) = Z$. The powers of W applied to Z therefore alternate between $1 - Z$ and Z, and $WAZ = \{(1 - Z) + \rho Z\}/(1 - \rho^2)$ lies in the span of the intercept and Z. The derivation of Equation 4 applies with the aliased spillover column dropped, $V = [1, Z, X_k]$ and $\theta = (\gamma, \tau - \gamma, \beta)^\top$, so that $\tau - \gamma$ takes the place of τ, and $g = -1/(1 + \rho)$. Two consequences follow: $\hat{\rho}$ is the same at every value of τ, and

$$\hat{\tau} - \tau = -\gamma + \delta + (\tau - \gamma)\,\frac{\hat{\rho} - \rho}{1 + \rho},$$

where δ collects the first two terms of that decomposition and, like $\hat{\rho}$, does not depend on τ. The error in $\hat{\rho}$ is therefore multiplied by $\tau - \gamma$, which grows with τ. The mean standard deviation of $\hat{\tau}$ rose from 0.202 at $\tau = 0.8$ to 0.327 at $\tau = 3$, and the standard error tracked it, rising on average from 0.199 to 0.322. With the bias held near $-\gamma$, wider intervals covered τ more often. The rise was largest at $\rho = 0$ (coverage 0.145 to 0.541) and smallest at $\rho = 0.5$ (0.130 to 0.265), consistent with the factor $1/(1 + \rho)$. The same term raised the MSE from 0.477 to 0.567. About three quarters of that increase was variance, $(n - 1)\mathrm{SD}^2/n$ averaged over scenarios (Appendix A2), which rose from 0.041 to 0.109. The other quarter was squared bias, which rose from 0.436 to 0.458 because $\hat{\rho}$ was biased slightly downward, moving the mean bias from -0.650 to -0.667. The higher coverage at large τ reflects wider intervals around the wrong quantity, not a recovery of τ.

Block-level detail. Under queen contiguity, the average rank of Incidence-Guided Saturation rose from 1.41 at $\tau = 0.8$ to 1.85 at $\tau = 3$, while that of Balanced Quartiles stayed between 2.13 and 2.14 and that of Isolation Buffer fell from 3.71 to 3.31. Incidence-Guided Saturation had the lower MSE of it and Balanced Quartiles in 103, 102, 102, 102, and 101 of the 160 blocks at $\tau = 0.8$, 1, 1.5, 2, and 3, and a lower MSE than Isolation Buffer in 75, 73, 56, 40, and 12 of the

80 control-only blocks. Its worst case across blocks was the lowest of the six designs at $\tau \leq 1.5$, but at $\tau = 2$ and $\tau = 3$ that of Isolation Buffer was lower (0.207 against 0.210, and 0.208 against 0.238). Under rook contiguity, at $\tau = 3$, Incidence-Guided Saturation had the lower MSE in 47 of the 80 control-only blocks, and its worst case across rook blocks was the lowest of the six designs at every value of τ except $\tau = 3$, where that of Isolation Buffer was lower (0.173 against 0.174). The coverage of rook Checkerboard was 0.138, 0.150, 0.198, 0.269, and 0.442 at $\tau = 0.8$, 1, 1.5, 2, and 3.

A8. Observed-incidence application detail

The primary application numbers come from the aggregate performance and allocation-risk summaries of the main run and of the separate tail refinement. Annual tables average the six queen parameter pairs equally, without pooled Monte Carlo intervals. Each random draw used a fixed, reproducible seed. An independent check recomputed performance from the stored fits, checked annual treatment budgets and regional orderings, and checked the allocations and outcomes reused from the main run in the tail refinement. The county death-certificate source data are restricted and not publicly available; only 58-cluster summaries are reported.

For randomized designs, the prespecified refinement schedule allowed $J = 100, 200, 400$ and $R = 100, 200, 400, 1000, 2000, 4000$; whichever source of Monte Carlo error, allocations or outcomes, contributed more to the shortfall from the precision targets was extended. All randomized designs met the targets at $J = 100$ and $R = 100$. For designs with a single possible assignment, only outcomes were extended. If an allocation occurs n_a times among J draws, its risk estimate enters the mean with weight n_a/J. The variance due to outcome noise therefore involves squared weights $(n_a/J)^2$, not n_a/J^2 as if duplicate estimates were independent. Bias and coverage use the same accounting of frequencies and covariances. The overall precision includes both variation across assignments and outcome noise given an assignment; a design whose sampled assignments happened to coincide is distinguished from one with only one possible assignment. Approximate

Monte Carlo standard errors of setting-specific differences between designs treat the designs' random draws as independent, whereas averages across settings or models do not assume that shared allocations are independent. Settings with the same set of possible assignments shared their simulations, with annual treatment counts and population shares recalculated.

Estimated risk variance is reported raw and with its outcome Monte Carlo noise contribution subtracted. Signed corrected estimates are retained rather than silently truncated; none is negative in the NC refinement. Quantiles and worst-decile means use noisy estimated conditional risks. To avoid selection bias, worst-decile assignments were identified with one half of the simulated outcomes and evaluated with the other half, and the roles of the two halves were then reversed. The Jaccard overlap of the two worst deciles describes sampled tail membership, not a confidence interval or population tail probability. For a design with only one possible assignment, an overlap of one is uninformative.

Design adaptations. Before simulation, we checked the 58-cluster row-standardized contiguity matrix W, its binary adjacency and its row order against the boundary geometry. On the NC map, Checkerboard visits clusters by decreasing degree, chooses the arm that conflicts least with assigned neighbors, breaks equal conflicts toward control, and complements the assignment if more than half are treated. In Balanced Quartiles and Balanced Halves, odd-size strata receive their necessary extra treatments at random; in Balanced Halves, one half receives 14 treatments and the other 15, with the extra-treatment half chosen randomly. The four-region partition used by both saturation designs was selected from 250 candidates generated by a geographic k-means procedure, by compactness and balance in reference population, cluster count and county count, without consulting outcomes; reference population is pooled person-years divided by four. Incidence-Guided Saturation first ranks clusters by annual per-capita rates, averages their ranks equally within each region, ranks those four means, and assigns 80/60/40/20% from highest to lowest. Regional ties are broken randomly per allocation, and local treatment

counts are rounded under the retained design rules. Per-capita cluster rates already account for population size; regional mean ranks answer an equally weighted cluster question, whereas pooled regional deaths/population weights larger clusters more heavily. Mean raw rates and pooled regional rates are evaluated as prespecified sensitivities rather than chosen by performance. The grid rule averages X directly: mean ranks align with its rank-scaled Poisson cases but are a deliberate adaptation for the application, rather than the exact rule used for continuous grid incidence. The 14 groups used by 2×2 Blocking are formed once per year by k-means on standardized projected coordinates, incidence ranks and log(1+population), then held fixed; half of each group, rounded, is treated at random, and the single-cluster group of each year remains control, because half of one cluster rounds to zero. This is not a set of literal square blocks.

The fixed regional criterion is $C = 3s_P + 1.5s_K + 1.5s_D + c/d_{\max}$, where s_P, s_K and s_D are the standard deviations of regional population, cluster and county shares. Here c averages within-region distances to regional centers and $d_{\max}$ is the maximum pairwise distance between the projected cluster coordinates. Candidates use standardized projected coordinates and are selected without outcome performance. The fixed groups used by 2×2 Blocking contain one to eight clusters; yearly total treatments are 28, 31, 27 and 29. Table 18 documents alternative summaries before performance evaluation.

Table 18: Regional ordering from highest to lowest summary, using the fixed region IDs in Figure 16. Primary mean ranks weight clusters equally; pooled regional rates weight cluster rates by population.

Year	Mean cluster rank	Mean cluster rate	Pooled regional rate
2018	2,1,3,4	2,4,1,3	4,3,2,1
2019	4,2,3,1	4,2,3,1	4,3,1,2
2020	2,4,1,3	2,4,1,3	4,3,2,1
2021	2,1,4,3	2,4,1,3	4,3,1,2

Allocation-risk estimates for the refined setting $\rho = 0.5$, $\gamma = 0.8$ (Figure 19) support the downside comparison in the body.

Table 19 retains MSE, bias and coverage for every design in every year; budgets are in Table 7.

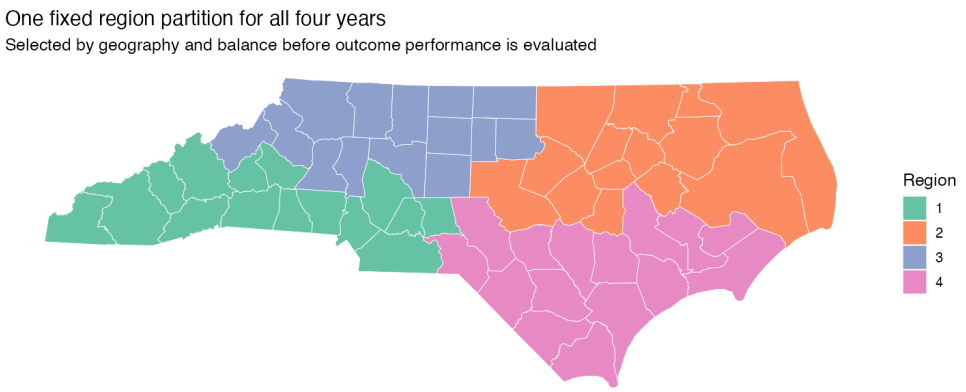


Figure 16: Fixed four-region partition used by both saturation strategies in all years, chosen from 250 geographic candidates before outcome evaluation.

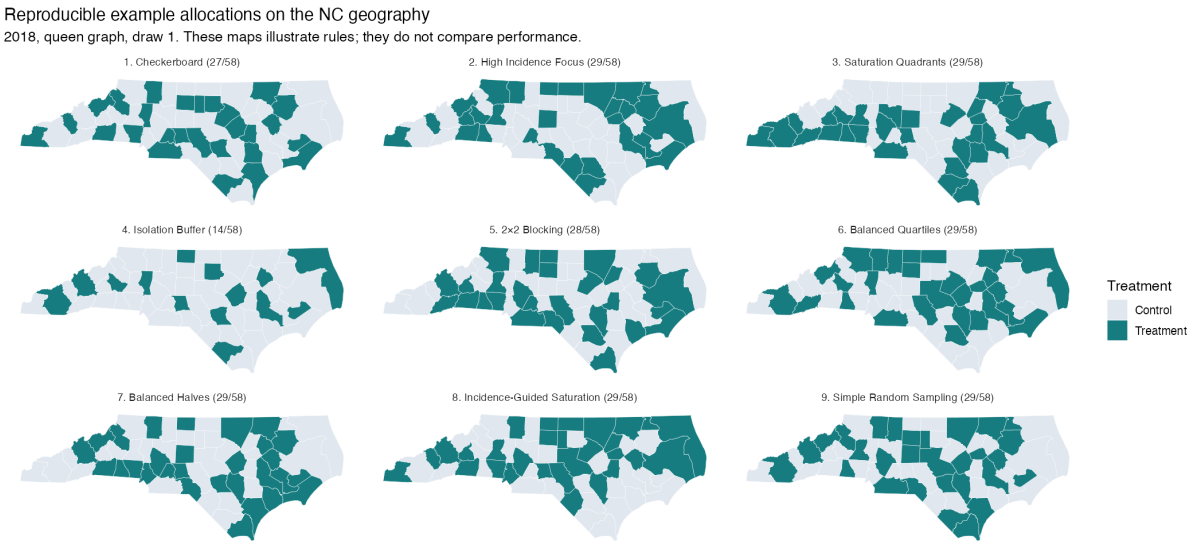


Figure 17: One illustrative queen assignment from each application strategy using 2018 observed incidence and fixed geography. These sampled assignments illustrate the adaptations; they were not selected by observed performance.

More granular setting-specific results, precision components, residual treatment information and warning and failure records are kept as aggregate output tables. The absence of warnings and failures is a verified result, not a sign that diagnostics were skipped.

Table 20 reports Incidence-Guided Saturation's matched annual mean-MSE ratios. Rook means use only the four extreme combinations of ρ and γ shared with the primary analysis. Baseline/X-adjusted and regional-summary means use the six queen pairs. These are matched contrasts,

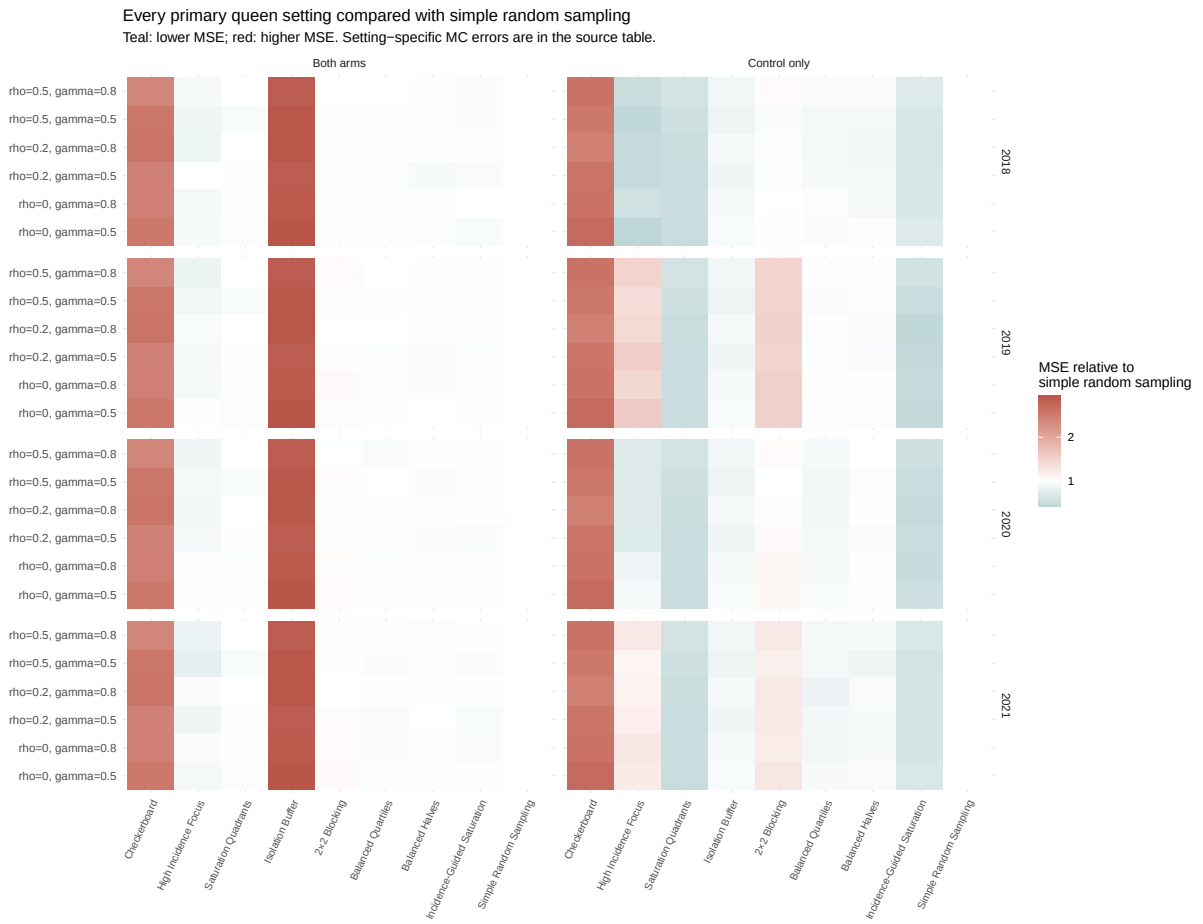


Figure 18: Primary queen setting-specific mean MSE ratios to year-matched SRS. Parameters and spillover regimes are kept separate; this plot does not pool neighbor definitions.

not ratios of unmatched pooled analyses.

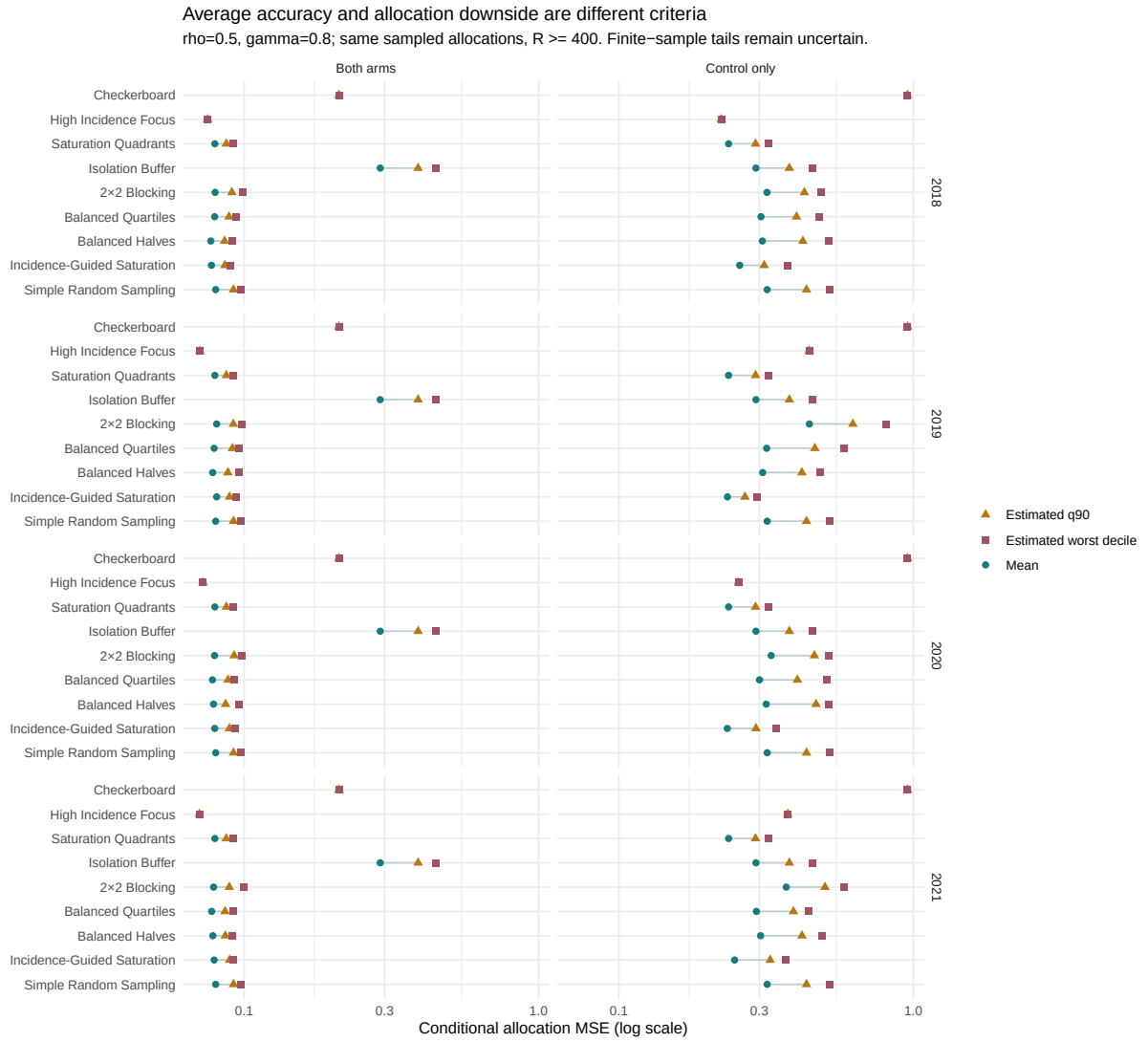


Figure 19: Allocation-specific conditional MSE in the refined primary queen setting $\rho = 0.5, \gamma = 0.8$, by year and spillover regime. Mean MSE, the estimated 90th percentile of allocation-specific MSE and the estimated worst-decile mean summarize the same $J = 100$ sampled assignments, with at least 400 independent outcomes per assignment of a randomized design or 1,000 for designs with a single possible assignment. These tail estimates, based on a finite number of outcomes, are descriptive, with uncertain membership; they are not quantiles of realized errors or a fresh allocation confirmation.

Table 19: Primary queen application performance by design and planning year. Each entry averages the six ρ/γ pairs of one spillover regime with equal weight; MSE, bias and coverage are Monte Carlo estimates. Checkerboard, Saturation Quadrants, Isolation Buffer and SRS use the same simulations in every year, because annual incidence enters neither their allocation nor the outcome model, so their rows repeat; repeated values are not independent annual confirmation. Treatment counts and population shares are in Table 7.

		Both arms			Control only		
Design	Year	MSE	Bias	Coverage	MSE	Bias	Coverage
Checkerboard	2018	0.21591	-0.0074	0.936	0.92036	0.0100	0.933
	2019	0.21591	-0.0074	0.936	0.92036	0.0100	0.933
	2020	0.21591	-0.0074	0.936	0.92036	0.0100	0.933
	2021	0.21591	-0.0074	0.936	0.92036	0.0100	0.933
High Incidence Focus	2018	0.07172	0.0160	0.934	0.21426	0.0237	0.932
	2019	0.07180	0.0063	0.936	0.43019	0.0227	0.931
	2020	0.07292	0.0094	0.931	0.25918	0.0327	0.930
	2021	0.07031	0.0104	0.934	0.35318	0.0296	0.929
Saturation Quadrants	2018	0.07518	0.0208	0.935	0.22329	0.0464	0.926
	2019	0.07518	0.0208	0.935	0.22329	0.0464	0.926
	2020	0.07518	0.0208	0.935	0.22329	0.0464	0.926
	2021	0.07518	0.0208	0.935	0.22329	0.0464	0.926
Isolation Buffer	2018	0.28591	0.0088	0.932	0.28575	0.0106	0.931
	2019	0.28591	0.0088	0.932	0.28575	0.0106	0.931
	2020	0.28591	0.0088	0.932	0.28575	0.0106	0.931
	2021	0.28591	0.0088	0.932	0.28575	0.0106	0.931
2×2 Blocking	2018	0.07609	0.0099	0.933	0.30622	0.0281	0.929
	2019	0.07808	0.0143	0.930	0.43630	0.0314	0.928
	2020	0.07764	0.0122	0.933	0.31848	0.0281	0.930
	2021	0.07736	0.0107	0.931	0.36168	0.0289	0.928
Balanced Quartiles	2018	0.07610	0.0170	0.933	0.29352	0.0315	0.929
	2019	0.07624	0.0175	0.933	0.30616	0.0367	0.929
	2020	0.07530	0.0158	0.933	0.28932	0.0356	0.927
	2021	0.07453	0.0175	0.934	0.28439	0.0354	0.929
Balanced Halves	2018	0.07458	0.0158	0.934	0.29101	0.0395	0.930
	2019	0.07517	0.0167	0.934	0.29955	0.0360	0.930
	2020	0.07466	0.0150	0.934	0.30610	0.0372	0.930
	2021	0.07584	0.0160	0.932	0.29089	0.0340	0.929
Incidence-Guided Saturation	2018	0.07428	0.0174	0.935	0.24285	0.0464	0.928
	2019	0.07628	0.0233	0.935	0.21565	0.0522	0.926
	2020	0.07682	0.0209	0.933	0.22029	0.0474	0.926
	2021	0.07450	0.0197	0.936	0.23732	0.0451	0.926
Simple Random Sampling	2018	0.07638	0.0135	0.932	0.30761	0.0378	0.929
	2019	0.07638	0.0135	0.932	0.30761	0.0378	0.929
	2020	0.07638	0.0135	0.932	0.30761	0.0378	0.929
	2021	0.07638	0.0135	0.932	0.30761	0.0378	0.929

Table 20: Incidence-Guided Saturation sensitivity MSE divided by matched primary MSE, by year and regime. Baseline adds rank-scaled X to both generation and fit; rook uses the matched extreme combinations of ρ and γ. Regional-summary alternatives preserve fixed regions.

Sensitivity	Regime	2018	2019	2020	2021
Baseline / adjusted	Both	1.016	1.030	1.015	1.035
Baseline / adjusted	Control	1.026	1.023	1.042	1.040
Rook contiguity	Both	1.005	0.998	0.985	0.998
Rook contiguity	Control	0.967	0.969	0.971	1.036
Mean regional rate	Both	1.034	1.000	1.000	1.031
Mean regional rate	Control	0.907	1.000	1.000	0.928
Pooled regional rate	Both	1.015	0.979	0.981	1.003
Pooled regional rate	Control	0.962	1.133	1.061	1.030